



Review

GeoNLU: Bridging the gap between natural language and spatial data infrastructures

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ABSTRACT

Integrating natural language processing (NLP) techniques with spatial data infrastructures (SDIs) potentially revolutionize the way users interact with geospatial data. This article presents GeoNLU, a comprehensive framework aimed at bridging the gap between natural language and SDIs. GeoNLU aims to enable seamless interaction and querying of geospatial data through natural language, thereby enhancing accessibility and usability for a wide range of users. This article delves into the theoretical foundations, architectural design, key components, and potential applications of GeoNLU, highlighting its significance in improving geospatial data exploration, analysis, and decision-making.

1. Introduction

1.1. Background and Motivation

Geospatial data plays a vital role in various domains, including urban planning, environmental management, and transportation. With the rapid advancement of technology and the availability of sophisticated data collection methods, the importance and complexity of geospatial data have increased significantly. This section provides an overview of the growing significance of geospatial data in these domains and highlights the challenges and opportunities associated with its management and utilization [1–5] and it is represented in Fig. 1.

1.1.1. Urban planning

Urban planning involves designing, developing, and managing cities to ensure efficient and sustainable use of resources [6]. Geospatial data provides valuable insights for urban planners in several ways. Geospatial data helps identify suitable areas for residential, commercial, and industrial development, considering factors such as infrastructure, environmental constraints, and population density. Geospatial data aids in analyzing transportation networks, optimizing routes, and identifying areas with high traffic congestion to improve transportation infrastructure and mobility. Geospatial data enables the monitoring of urban expansion, identifying growth patterns, and supporting informed decision-making for infrastructure development and resource

allocation.

1.1.2. Environmental management

Geospatial data plays a crucial role in understanding and managing the environment addressing various environmental challenges [7]. Geospatial data helps monitor and manage natural resources such as forests, water bodies, and agricultural land, enabling sustainable utilization and conservation. Geospatial data assists in disaster risk assessment, response planning, and post-disaster recovery efforts by providing information on vulnerable areas, evacuation routes, and resource allocation. Geospatial data supports the assessment of climate change impacts, such as sea-level rise, temperature changes, and ecosystem shifts, enabling effective adaptation and mitigation strategies.

1.1.3. Transportation

Geospatial data plays a critical role in optimizing transportation systems and improving mobility [8]. Geospatial data provides accurate and up-to-date information about roads, traffic conditions, and alternative routes, facilitating efficient navigation and reducing travel times. Geospatial data is utilized in intelligent transportation systems for real-time traffic monitoring, congestion management, and signal optimization, improving overall traffic flow and reducing emissions. Geospatial data enables the provision of location-based services, such as ride-sharing, delivery tracking, and public transportation information, enhancing convenience and efficiency.

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The increasing importance and complexity of geospatial data in these domains pose several challenges, including data integration from diverse sources, data quality assurance, interoperability, and efficient data processing and analysis. However, these challenges also present opportunities for developing advanced technologies, such as integrating natural language processing with spatial data infrastructures, as proposed in the GeoNLU framework. By bridging the gap between natural language and geospatial data, GeoNLU aims to enhance accessibility and usability, enabling users to interact with geospatial data more effectively and supporting informed decision-making in various domains.

1.2. Challenges faced by users in interacting with geospatial data

Interacting with geospatial data poses several challenges for users, including the need for specialized technical skills and complex query languages. These challenges limit the accessibility and usability of geospatial data, hindering effective decision-making and analysis.

1.2.1. Technical skills requirement

Geospatial data often requires specialized technical skills to understand, manipulate, and interpret. Some challenges include working with geospatial data often requires proficiency in GIS software, which have a steep learning curve for non-technical users [9]. Performing operations like spatial queries, data fusion, and geoprocessing require knowledge of specialized tools and programming languages [10]. Integrating data from various sources with different formats and coordinate systems are complex, requiring expertise in data transformation and integration techniques [11].

1.2.2. Complex query languages

Geospatial data is typically queried using specific query languages, such as Structured Query Language (SQL) [12] or Spatial Query Language (SpatQL) [13]. This poses challenges for users. Query languages often have complex syntax and require users to understand the underlying data structure and spatial concepts [14]. Users without a technical background struggle to formulate and execute queries, impeding their ability to retrieve and analyze geospatial data effectively. Querying

geospatial data using traditional query languages do not provide user-friendly interfaces, making it challenging for non-experts to interact with the data.

1.2.3. Data complexity and volume

Geospatial data are voluminous and complex, posing user challenges in data management and processing. Geospatial data, especially in raster format, are massive in size, making it challenging to handle and analyze using standard hardware and software. Geospatial data comes from various sources and formats, requiring users to navigate through different data structures, coordinate systems, and metadata. Geospatial data often contains errors, inaccuracies, and uncertainties that need to be considered during analysis and decision-making.

1.2.4. Domain-specific knowledge

Interpreting and analyzing geospatial data effectively often requires domain-specific knowledge and expertise. Interpreting geospatial data requires an understanding of spatial relationships, proximity analysis, and spatial patterns specific to the domain [15]. Geospatial data analysis often requires integrating spatial information with non-spatial attributes and contextual knowledge to derive meaningful insights.

Addressing these challenges is crucial to democratize geospatial data and make it accessible to a broader range of users. The proposed GeoNLU framework aims to bridge the gap between natural language and geospatial data, providing a more intuitive and accessible interface for interacting with geospatial information. By enabling users to use natural language queries and reducing the reliance on complex query languages, GeoNLU seeks to enhance the accessibility and usability of geospatial data, empowering a wider audience to leverage its potential for decision-making and analysis.

1.3. Limitations of existing approaches to geospatial data interaction

Existing approaches to geospatial data interaction have limitations that impede the effective utilization and accessibility of geospatial information. Some of the key limitations include the following.



Fig. 1. Overview of the growing significance of geospatial data.

1.3.1. Reliance on traditional query languages

Many geospatial data systems rely on traditional query languages like Structured Query Language (SQL) or specialized spatial query languages. These languages often have complex syntax and require users to possess technical expertise in writing precise and accurate queries. This reliance on query languages poses challenges for non-technical users or those without a background in geospatial data analysis, limiting their ability to interact with and extract insights from geospatial data [16].

1.3.2. Lack of intuitive interfaces

Geospatial data systems often lack intuitive interfaces for data interaction. Traditional interfaces, such as command-line interfaces or complex GUIs, are overwhelming for users unfamiliar with geospatial concepts or who do not possess technical skills. This limitation makes it difficult for non-experts to effectively interact with and explore geospatial data, hindering their ability to leverage the full potential of geospatial information for decision-making and analysis.

1.3.3. Limited natural language interaction

Most existing geospatial data systems do not provide robust support for natural language interaction. Users typically have to rely on predefined query structures or predefined queries, restricting their ability to express complex queries or explore data using their own language. This limitation hinders the accessibility and usability of geospatial data for a broader range of users who do not possess the technical knowledge or familiarity with query languages.

1.3.4. Lack of contextual understanding

Existing geospatial data systems often struggle to capture and interpret contextual information. Geospatial data is inherently spatial, and understanding it requires considering the spatial relationships, context, and domain-specific knowledge. However, many systems do not effectively incorporate contextual understanding, leading to limitations in analyzing and interpreting geospatial data.

1.3.5. Scalability and performance

Geospatial data systems face challenges in handling large-scale datasets or performing complex spatial operations efficiently. Processing and analyzing vast volumes of geospatial data in real-time are computationally intensive and require specialized infrastructure and optimization techniques. Inadequate scalability and performance limit the effectiveness of geospatial data interaction, particularly when dealing with big data or time-sensitive applications.

Addressing these limitations is essential to enhance the accessibility, usability, and effectiveness of geospatial data interaction. The proposed GeoNLU framework aims to overcome these limitations by leveraging natural language processing techniques to bridge the gap between natural language and geospatial data. By enabling users to interact with geospatial data using familiar language and intuitive interfaces, GeoNLU strives to democratize geospatial data access and empower a broader range of users to utilize geospatial information for various applications and decision-making processes effectively.

1.4. The need for bridging natural language and SDIs

Integrating Natural Language Processing (NLP) techniques with Spatial Data Infrastructures (SDIs) holds immense importance in promoting accessibility and democratizing geospatial information [17]. Here are several justifications for the significance of this integration.

1.4.1. Enhanced accessibility

By enabling users to interact with geospatial data using natural language, the integration of NLP with SDIs removes barriers for non-technical users or those unfamiliar with complex query languages. Natural language queries provide a more intuitive and familiar way of interacting with data, making geospatial information accessible to a

broader range of users. This accessibility promotes inclusivity and allows individuals from various backgrounds and expertise levels to leverage geospatial data for decision-making and analysis.

1.4.2. User-friendly interface

NLP integration with SDIs facilitates the development of user-friendly interfaces that are more intuitive and easier to use. Natural language queries are closer to how people naturally communicate and express their information needs, eliminating the need for users to learn complex query languages or data manipulation techniques. This user-friendly interface lowers the learning curve, enhances user experience, and encourages wider adoption of geospatial data tools and platforms.

1.4.3. Democratizing geospatial information

Integrating SDIs with NLP is crucial in democratizing geospatial information [18]. Traditional geospatial data interaction methods often require specialized technical skills, limiting access to a small group of experts. By enabling natural language interactions, individuals without extensive geospatial or technical backgrounds effectively engage with geospatial data. This democratization empowers a broader range of users, including policymakers, researchers, and the general public, to utilize geospatial information for informed decision-making, research, and community engagement.

1.4.4. Reduced dependency on query languages

Many existing SDIs rely on query languages, which is challenging for non-experts or casual users to master. NLP integration reduces the dependency on complex query languages and allows users to express their information needs in everyday language. This approach enables users to focus on the content and meaning of their queries rather than the syntax and structure of a specific query language. It empowers users to retrieve and analyze geospatial data without requiring extensive training or expertise in query languages [19].

1.4.5. Natural language understanding

NLP techniques go beyond simple keyword matching and incorporate advanced natural language understanding capabilities [20]. These techniques enable the system to interpret user queries, understand context, and provide meaningful responses. By leveraging natural language understanding, SDIs deliver more accurate and relevant results, enhancing the user experience and enabling users to extract valuable insights from geospatial data more effectively.

1.4.6. Bridging the gap between experts and non-experts

The integration of NLP with SDIs helps bridge the gap between geospatial experts and non-experts. Experts focus on domain-specific analysis and decision-making tasks, while non-experts still access and explore geospatial data using natural language queries. This collaboration between experts and non-experts fosters collaboration and knowledge sharing and facilitates a wider range of perspectives in utilizing geospatial data [21].

Thus, integrating NLP techniques with SDIs to enable natural language interaction with geospatial data is crucial for promoting accessibility, democratizing geospatial information, and bridging the gap between technical experts and non-experts. This integration empowers users with diverse backgrounds and levels of expertise to engage with geospatial data, facilitating informed decision-making, research, and societal engagement with geospatial information.

Section 2 discusses the literature review, while Section 3 provides an overview of the theoretical foundations of the article. The GEONLU framework is proposed in Section 4, and Section 5 represents the Query Processing and Analysis. The Spatial Knowledge Graph is discussed in Section 6, and the user interface and interaction are discussed in Section 7. Section 8 discusses applications and use cases, while Section 9 addresses challenges and future directions.

2. Literature review

2.1. Overview of geospatial NLP research

Geospatial Natural Language Processing (NLP) is a rapidly growing field that combines the techniques of NLP with geographic information systems (GIS) to extract spatial information from text. This information then be used for a variety of purposes. Geocoding is the process of converting text descriptions of places into their corresponding geographic coordinates. This is used to create maps of places mentioned in the text or to identify the locations of events or activities [22]. Place recognition is the process of identifying the places mentioned in the text. This is used to track the movement of people or objects, or to identify the locations of businesses or other organizations [23]. Event detection is the process of identifying events mentioned in the text. This is used to track the occurrence of natural disasters, or to identify the locations of protests or other social events [24]. Sentiment analysis is the process of identifying the sentiment of text, such as whether it is positive, negative, or neutral. This is used to track public opinion about a particular topic, or to identify the impact of a marketing campaign [25]. Geospatial NLP is a challenging field, due to the complexity of natural language and the diversity of geographic data. However, it is also a very promising field, with the potential to revolutionize the way we interact with geographic information. The overview of the research is represented in Fig. 2.

2.1.1. Methodologies

There are a variety of methodologies that are used for geospatial NLP. One common approach is using a pipeline of NLP techniques, each performing a specific task. For example, the first technique is used to identify place names, the second technique is used to geocode those place names, and the third technique is used to identify the sentiment of the text.

Another approach is to use a single NLP technique designed explicitly for geospatial data. For example, there are several NLP techniques that are used to identify the spatial relationships between entities mentioned in the text.

The choice of methodology will depend on the specific task at hand. For example, if the task is to geocode a large corpus of text, then a pipeline of NLP techniques is the best approach. However, if the task is to identify the spatial relationships between entities mentioned in the text, then a single NLP technique is the best approach.

2.1.2. Techniques

There are a variety of NLP techniques that are used for geospatial NLP. Some of the most common techniques includes the following. Part-of-speech tagging is the process of identifying the part of speech of each

word in a sentence. This information is used to identify place names, events, and other entities mentioned in [25]. Named entity recognition is the process of identifying named entities in the text, such as people, organizations, and places. This information is used to geocode text and to identify the locations of events and activities [26]. Coreference resolution is the process of identifying the referents of pronouns and other words that refer to entities mentioned in the text [27]. This information is used to improve the accuracy of geocoding and place recognition [28]. Spatial reasoning is the process of using spatial knowledge to reason about the relationships between entities mentioned in text. This information is used to identify the spatial relationships between entities, such as the distance between two places or the direction of travel between two points [29].

2.1.3. Applications

Geospatial NLP has a wide range of applications. Geospatial NLP is used to create maps of places mentioned in text. This is used to visualize the distribution of places mentioned in a corpus of the text, or to track the movement of people or objects [30]. Geospatial NLP is used to detect events mentioned in the e-text. This is used to track the occurrence of natural disasters, or to identify the locations of protests or other social events [31]. Geospatial NLP is used to identify the sentiment of text about a particular place. This is used to track public opinion about a particular place, or to identify the impact of a marketing campaign [32]. Geospatial NLP is used to gain insights into the spatial patterns of human behavior. This information is used to improve business decisions, such as where to open a new store or how to target a marketing campaign [33]. Geospatial NLP is a rapidly growing field with a wide range of applications. As the field continues to develop, we expect to see even more innovative and powerful applications of geospatial NLP.

2.2. Existing approaches to integrating NLP with SDIs

Connecting Natural Language Processing (NLP) techniques with Spatial Data Infrastructures (SDIs) involves integrating language understanding capabilities with geospatial data systems. Several methods are employed to facilitate this connection, including semantic annotation, ontology-based approaches, and information extraction, illustrated in Fig. 3.

2.2.1. Semantic annotation

Semantic annotation involves enriching geospatial data with semantic meaning by associating textual descriptions or labels with specific geographic features. In the context of connecting NLP with SDIs, semantic annotation is used to bridge the gap between natural language expressions and geospatial concepts. This method annotates geospatial

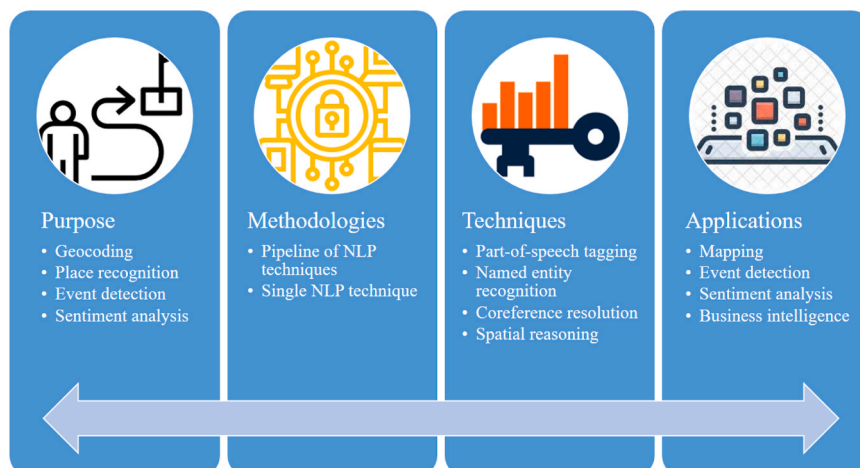


Fig. 2. Overview of the Geospatial NLP Research.

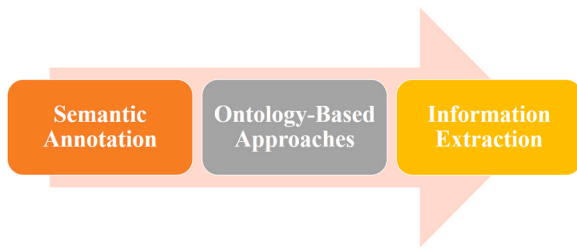


Fig. 3. Methods to integrate NLP with SDIs.

data with relevant semantic information, such as named entities, spatial relations, or thematic attributes, to better understand and interpret [34, 35].

For example, a geospatial dataset is annotated with semantic tags to identify and extract specific locations mentioned in natural language queries or texts. This enables the system to understand and process user queries more contextually meaningfully, improving the effectiveness of geospatial NLP.

2.2.2. Ontology-based approaches

Ontology-based approaches involve using formal knowledge representation frameworks to model a particular domain's concepts, relationships, and semantics. In the context of geospatial NLP, ontologies provide a structured representation of geospatial knowledge, facilitating the understanding and interpretation of natural language expressions [36,37].

By incorporating ontologies, SDIs capture and represent the spatial and semantic relationships between geospatial entities and concepts. Ontologies help in mapping natural language expressions to specific geospatial entities, enabling effective query understanding and data retrieval. Ontology-based approaches also support reasoning and inference capabilities, allowing the system to infer additional knowledge or make deductions based on the available geospatial information.

2.2.3. Information extraction

Information extraction techniques are used to identify and extract relevant information from unstructured or semi-structured text sources. In the context of connecting NLP with SDIs, information extraction methods are employed to extract geospatial information from natural language texts, such as documents, social media posts, or web pages [38].

Geospatial information extraction involves techniques like named entity recognition, spatial relation extraction, and geoparsing. Named entity recognition identifies and classifies location names, addresses, or other geospatial references in text. Spatial relation extraction focuses on identifying spatial relationships between geospatial entities, such as proximity or containment. Geoparsing involves parsing and extracting location-based information from text, such as coordinates or administrative divisions.

By applying information extraction techniques, geospatial NLP systems automatically extract relevant geospatial information from textual data, facilitating the integration of natural language expressions with SDIs.

These methods, namely semantic annotation, ontology-based approaches, and information extraction, play vital roles in connecting NLP techniques with SDIs. They enable the interpretation of natural language expressions, facilitate query understanding, and enhance the interaction and integration of geospatial data with language-based interfaces. By leveraging these methods, geospatial NLP systems bridge the gap between textual information and spatial data, enabling seamless and intuitive interaction with geospatial information.

2.3. Limitations and gaps in current methodologies

Existing approaches in connecting NLP techniques with SDIs face several limitations and challenges that hinder their effectiveness and scalability, as shown in Fig. 4.

2.3.1. Lack of semantic understanding

One significant limitation is the lack of deep semantic understanding in existing approaches. While NLP techniques process and analyze text, understanding natural language expressions' complex semantics and contextual meanings remains challenging. Geospatial NLP systems struggle to accurately interpret nuanced queries or handle ambiguous language, leading to incorrect or incomplete results. Enhancing the semantic understanding capabilities of these systems is crucial to improve accuracy and user satisfaction [39].

2.3.2. Scalability issues

Many existing approaches face scalability issues when dealing with large volumes of geospatial data. Geospatial datasets are vast and complex, and processing natural language queries over such datasets are computationally intensive. Scaling up the systems to handle high volumes of data and user queries in real time is a significant challenge. Efficient indexing, storage, and retrieval mechanisms need to be developed to ensure optimal performance and scalability of geospatial NLP systems [40].

2.3.3. Efficient query processing

Efficient query processing is a crucial challenge in geospatial NLP. Traditional query processing methods do not well-suited for handling complex natural language queries over geospatial data. Query optimization, including query rewriting and query execution planning, is essential to ensure efficient retrieval of geospatial information. Developing innovative techniques to process natural language queries and optimize geospatial query execution is vital for improving system performance and responsiveness [41].

2.3.4. Data integration and heterogeneity

Geospatial data often comes from diverse sources and exhibit heterogeneity in terms of data formats, schemas, and spatial reference systems. Integrating and harmonizing such diverse geospatial data sources with natural language interfaces are challenging. Semantic



Fig. 4. Limitations and Gaps in current methodologies.

mapping, data fusion, and data interoperability techniques are necessary to address these challenges and enable seamless integration and retrieval of geospatial information [42].

2.3.5. Limited training data and domain-specific knowledge

Training effective NLP models for geospatial applications requires substantial amounts of labeled training data. However, acquiring large-scale annotated geospatial datasets are costly and time-consuming. Additionally, geospatial domains have specific terminologies and concepts that are not adequately represented in general-purpose language models. Building domain-specific knowledge bases and leveraging transfer learning techniques help overcome these challenges and improve the performance of geospatial NLP systems [43].

2.3.6. Privacy and security concerns

Geospatial data often contains sensitive information related to individuals, organizations, or critical infrastructure. Ensuring privacy and security when integrating NLP techniques with SDIs is crucial. Safeguarding personally identifiable information, secure data access controls, and protecting against unauthorized data disclosure are vital considerations for geospatial NLP systems [44].

Addressing these limitations and challenges requires ongoing research and innovation in geospatial NLP. Advancements in semantic understanding, scalable processing techniques, efficient query optimization, data integration methods, and privacy protection mechanisms will contribute to the development of more effective and robust geospatial NLP systems.

3. Theoretical foundations

3.1. Natural Language Understanding (NLU) techniques and algorithms

In GeoNLU, various Natural Language Understanding (NLU) techniques are employed to bridge the gap between natural language expressions and geospatial data. These techniques include rule-based approaches, statistical models, and machine learning algorithms, as shown in Fig. 5.

3.1.1. Rule-based approaches

Rule-based approaches in GeoNLU involve using predefined rules or patterns to extract relevant geospatial information from natural language expressions. These rules are typically crafted by domain experts and linguists and capture specific syntactic or semantic patterns that indicate geospatial references or queries. Rule-based approaches are often effective in handling simple and well-defined geospatial queries, where the patterns are easily defined and matched. However, they struggle with handling complex or ambiguous queries and require manual rule creation and maintenance [45].

3.1.2. Statistical models

Statistical models are widely used in GeoNLU to extract geospatial information from text by leveraging statistical patterns and probabilistic algorithms. These models analyze large volumes of textual data and learn statistical relationships between words, phrases, and geospatial entities. Techniques like Named Entity Recognition (NER) and Part-of-

Speech (POS) tagging are commonly employed in statistical models in GeoNLU. These models are trained on annotated datasets, allowing them to identify and extract geospatial entities, such as location names, addresses, or coordinates, based on statistical patterns observed in the training data [46].

3.1.3. Machine learning algorithms

Machine learning algorithms play a crucial role in GeoNLU, enabling systems to automatically learn patterns and relationships between natural language expressions and geospatial data. Supervised learning algorithms, such as Support Vector Machines (SVM), Random Forests, or Neural Networks, are commonly used in GeoNLU tasks. These algorithms are trained on labeled datasets, where natural language expressions are annotated with corresponding geospatial information. The trained models then be used to predict geospatial information for new, unseen natural language queries or texts. Machine learning algorithms offer flexibility, scalability, and the potential to handle complex geospatial queries. However, they require substantial amounts of labeled training data and appropriate feature engineering to achieve optimal performance [47].

GeoNLU often combines multiple NLU techniques to leverage their respective strengths. For example, rule-based approaches are used as initial filters or for handling specific cases, while statistical models and machine learning algorithms provide more generalizable and robust geospatial information extraction capabilities.

It's important to note that the choice of NLU techniques in GeoNLU depends on factors such as the complexity of geospatial queries, the availability of labeled training data, and the specific requirements of the application domain. The combination of rule-based, statistical, and machine learning approaches provides a comprehensive toolkit for extracting and understanding geospatial information from natural language expressions.

3.2. Geospatial data representation and semantics

Geospatial data is represented using different models, including vector and raster formats. These models capture different aspects of geospatial information and serve various purposes in geospatial understanding. Additionally, semantics play a crucial role in geospatial understanding by providing context and meaning to the spatial data. This is represented in Fig. 6.

3.2.1. Vector representation

Vector representation is a widely used model for geospatial data. This model represents geographic features as discrete objects with precise geometric shapes (points, lines, polygons) and attributes. Each object is defined by its coordinates and associated attributes such as name, population, or land use. Vector data is versatile and suited for representing discrete objects, such as roads, buildings, or administrative boundaries. It enables accurate geometric representation, topological relationships, and attribute information. Vector data is commonly used in applications like urban planning, transportation, and land management [48].



Fig. 5. NLU Techniques.

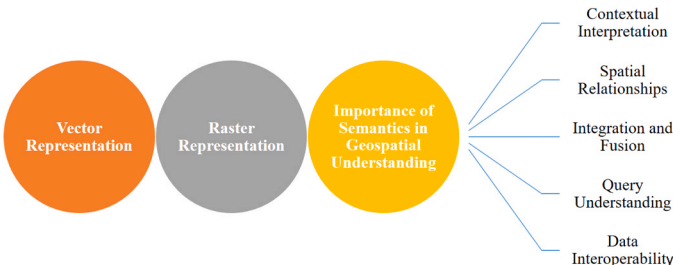


Fig. 6. Geospatial Data Representation and Semantics.

3.2.2. Raster representation

Raster representation uses a grid-like structure to represent geospatial data. In this model, the geographic space is divided into cells or pixels, and each cell stores a value representing a particular attribute or measurement. Raster data is well-suited for continuous phenomena like elevation, temperature, or satellite imagery. It provides a regular grid structure and allows for analysis based on cell values, such as interpolation, terrain analysis, or image processing. Raster data is widely used in applications like remote sensing, environmental modeling, and natural resource management [49].

3.2.3. Importance of semantics in geospatial understanding

Semantics play a critical role in geospatial understanding by providing context, meaning, and relationships to the spatial data. Geospatial data alone contain coordinates, shapes, or values, but semantics add layers of interpretation and understanding. Here are a few key reasons why semantics are important.

a. Contextual Interpretation: Semantics provide the necessary context to interpret geospatial data. For example, understanding that a specific coordinate represents a city's location or a polygon represents a park requires semantic information about the types and meanings of these entities [50].

b. Spatial Relationships: Semantics help in capturing and interpreting spatial relationships between geospatial entities. For example, knowing that a building is adjacent to a road or a river flowing through a particular region adds valuable information for analysis and decision-making [51].

c. Integration and Fusion: Semantics facilitate the integration and fusion of diverse geospatial data sources. By capturing the semantic meaning of attributes or concepts, data from different sources are harmonized and analyzed together, enabling comprehensive geospatial understanding [52].

d. Query Understanding: Semantics aid in understanding natural language queries or expressions related to geospatial data. By incorporating semantic knowledge, geospatial NLP systems better interpret and respond to user queries, improving the user experience and retrieval accuracy [53].

e. Data Interoperability: Semantics support data interoperability by providing common vocabularies and standards for geospatial data exchange. Semantically annotated data is easily understood and integrated across different systems or domains [54].

Vector and raster models offer distinct representations of geospatial data, capturing different types of spatial and attribute information. Semantics add context, meaning, and relationships to geospatial data, enhancing geospatial understanding, integration, and analysis. Combining appropriate data representation models and semantic knowledge is crucial for effective geospatial applications and decision-making.

3.3. Semantic web technologies for geospatial data integration

Semantic web technologies are vital in integrating geospatial data and enabling interoperability by providing standards and frameworks for representing, querying, and reasoning about data. Three key technologies in the semantic web stack are RDF (Resource Description Framework), OWL (Web Ontology Language), and SPARQL (SPARQL Protocol and RDF Query Language).

3.3.1. RDF (Resource Description Framework)

RDF is a foundational technology in the semantic web that provides a flexible data model for representing knowledge in a machine-readable format. RDF represents information as triples, consisting of subject-predicate-object statements, which form a graph-based data structure. RDF allows for expressing relationships, attributes, and metadata about resources. In the context of geospatial data, RDF represents geospatial entities, their attributes, spatial relationships, and other relevant

information. RDF provides a uniform way to structure and link geospatial data, facilitating integration across different sources and domains [55].

3.3.2. OWL (Web Ontology Language)

OWL is an ontology language in the semantic web that allows the formal specification and modeling of a domain's concepts, relationships, and constraints. OWL provides a rich vocabulary and expressive constructs to define ontologies, which capture the semantics of geospatial data. Ontologies define classes, properties, and axioms that represent the meaning and relationships between geospatial entities. By leveraging OWL, geospatial ontologies are developed, enabling the integration, and sharing of geospatial knowledge. OWL supports reasoning capabilities, allowing for automatic inference and deduction based on the defined ontological knowledge [56].

3.3.3. SPARQL (SPARQL protocol and RDF query language)

SPARQL is a query language designed for querying RDF data. It provides a standardized way to retrieve and manipulate RDF-based data. SPARQL enables complex queries over geospatial data by combining graph pattern matching, filtering, aggregations, and optional pattern matching. With SPARQL, geospatial data is queried to retrieve specific information, perform spatial and attribute-based filtering, and obtain insights from interconnected data sources. SPARQL allows for federated queries, where geospatial data from multiple distributed sources are queried together, promoting interoperability and data integration [57].

By leveraging RDF, OWL, and SPARQL, geospatial data is represented, linked, queried, and reasoned about in a standardized and interoperable manner. These technologies enable the integration of geospatial data from diverse sources and facilitate seamless data exchange and interoperability between geospatial systems. By adopting semantic web technologies, geospatial applications benefit from increased data interoperability, improved data integration, and enhanced geospatial analysis and decision-making capabilities.

4. GEONLU framework

4.1. Architectural overview

The GeoNLU framework is designed to bridge the gap between natural language expressions and geospatial data by leveraging Natural Language Understanding (NLU) techniques and spatial data infrastructures (SDIs). It follows a modular architecture that incorporates various components to enable seamless interaction with geospatial data. The detailed architectural overview of the GeoNLU framework and the interactions between its components is shown in Fig. 7.

4.1.1. User interface

The User Interface component serves as the entry point for users to interact with the GeoNLU framework. It provides a user-friendly interface where users input natural language queries or expressions related to geospatial data. The User Interface component captures user inputs and passes them to the NLU Engine for processing.

4.1.2. NLU engine

The NLU Engine is the core component of the GeoNLU framework, responsible for understanding and extracting geospatial information from natural language inputs. It employs various NLU techniques, such as rule-based approaches, statistical models, and machine learning algorithms, to analyze and interpret user queries. The NLU Engine applies techniques like Named Entity Recognition (NER), Part-of-Speech (POS) tagging, and semantic parsing to extract geospatial entities, spatial relationships, and relevant attributes from the natural language expressions.

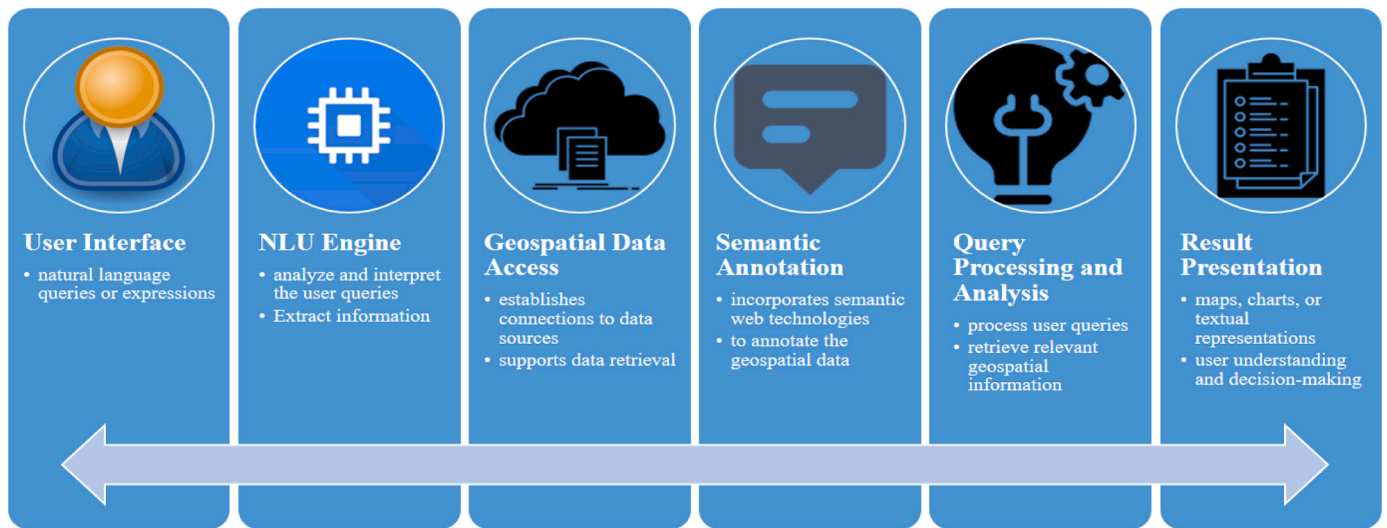


Fig. 7. Architectural Overview of the proposed system.

4.1.3. Geospatial data access

The Geospatial Data Access component provides connectivity to spatial data infrastructures (SDIs) and geospatial databases. It establishes connections to data sources, such as geographic information systems (GIS), geospatial web services, or geodatabases, to retrieve relevant geospatial data. This component supports data retrieval based on the extracted geospatial information from the NLU Engine, such as querying for specific locations, spatial features, or attribute values.

4.1.4. Semantic annotation

The Semantic Annotation component enriches the geospatial data with semantic information. It incorporates semantic web technologies, such as RDF (Resource Description Framework) and OWL (Web Ontology Language), to annotate the geospatial data with standardized vocabularies, ontologies, and semantic relationships. Semantic annotation enhances the interoperability and understanding of geospatial data by capturing the meaning, context, and relationships associated with the data.

4.1.5. Query processing and analysis

The Query Processing and Analysis component handles the processing and analysis of user queries and geospatial data. It utilizes techniques like spatial indexing, spatial analysis, and spatial reasoning to process user queries and retrieve relevant geospatial information efficiently. This component supports functionalities like spatial filtering, spatial joins, spatial aggregations, and spatial reasoning to provide meaningful insights and answers to user queries.

4.1.6. Result presentation

The Result Presentation component is responsible for presenting the retrieved geospatial information and analysis results to the user. It formats and visualizes the data in a user-friendly manner, such as maps, charts, or textual representations. The Result Presentation component ensures that the geospatial information is presented in a meaningful and intuitive way to facilitate user understanding and decision-making.

The interactions between these components in the GeoNLU framework involve a seamless flow of data and information. The User Interface component captures user inputs and passes them to the NLU Engine. The NLU Engine processes the natural language expressions, extracts geospatial information, and passes it to the Geospatial Data Access component for retrieving relevant geospatial data. The Semantic Annotation component enhances the geospatial data with semantic information. The Query Processing and Analysis component processes user queries and geospatial data, applying spatial operations and

analysis. Finally, the Result Presentation component presents the retrieved geospatial information and analysis results back to the user through the User Interface.

The modular design of the GeoNLU framework enables the integration of NLU techniques with spatial data infrastructures, facilitating seamless interaction with geospatial data and promoting accessibility and usability for users.

4.2. Core components and their functionalities

The GeoNLU framework comprises several key components that work together to bridge the gap between natural language expressions and geospatial data.

4.2.1. Natural language understanding (NLU) Module

The NLU module is responsible for understanding and interpreting natural language expressions related to geospatial data. It employs various NLU techniques, such as rule-based approaches, statistical models, and machine learning algorithms, to analyze and extract geospatial entities, spatial relationships, and relevant attributes from user queries. This module applies techniques like Named Entity Recognition (NER), Part-of-Speech (POS) tagging, and semantic parsing to process and understand the user's natural language inputs.

4.2.2. Data integration component

The data integration component focuses on integrating diverse geospatial data sources into a unified framework. It establishes connections with spatial data infrastructures (SDIs), geospatial databases, and other relevant data sources. This component enables the retrieval of geospatial data based on user queries and provides a seamless interface for accessing and managing geospatial datasets. It ensures that the necessary geospatial data is available for further processing and analysis.

4.2.3. Spatial knowledge graph

The spatial knowledge graph is a critical component of GeoNLU that captures and represents geospatial data's semantic relationships and spatial context. It utilizes semantic web technologies, such as RDF (Resource Description Framework) and OWL (Web Ontology Language), to create a graph-based representation of geospatial entities, their attributes, and spatial relationships. The spatial knowledge graph enhances the interoperability and understanding of geospatial data by providing a structured and standardized representation that is queried and reasoned upon.

4.2.4. Query processing and analysis module

The Query Processing and Analysis Module handles the processing and analysis of user queries and geospatial data. It applies spatial operations, such as spatial filtering, spatial joins, and spatial aggregations, to retrieve relevant geospatial information based on user queries. This module leverages spatial indexing techniques and spatial analysis algorithms to efficiently process large volumes of geospatial data and provide meaningful insights and answers to user queries. It enables spatial reasoning and analysis to support complex geospatial queries and decision-making processes.

4.2.5. Visualization and interaction component

The visualization and interaction component focuses on presenting the retrieved geospatial information and analysis results to users intuitively and interactively. It employs visualization methods such as maps, charts, and graphical depictions to represent geospatial data and analysis outputs visually. This component enables users to interact with the visualizations, explore different aspects of the data, and gain insights from the geospatial information. It provides a user-friendly interface for researchers to interact with the geospatial data and make informed decisions based on the presented information.

These key components of GeoNLU work together to enable seamless interaction with geospatial data using natural language expressions. The NLU module understands user queries, the data integration component retrieves relevant geospatial data, the spatial knowledge graph captures the semantic relationships, the query processing and analysis module performs spatial operations and analysis, and the visualization and interaction component present the results to users in a user-friendly manner. This integrated approach facilitates accessibility, usability, and effective utilization of geospatial data for decision-making processes.

4.3. Integration with SDIs and data sources

GeoNLU seamlessly integrates with different Spatial Data Infrastructures (SDIs) and data sources by employing various techniques and protocols to ensure compatibility and efficient data retrieval. Following are some key aspects of how GeoNLU achieves this integration.

4.3.1. Standardized protocols

GeoNLU utilizes standardized protocols and interfaces to connect with SDIs and data sources. It supports protocols such as OGC (Open Geospatial Consortium) standards [58], including WMS (Web Map Service) [59], WFS (Web Feature Service) [60], and WCS (Web Coverage Service) [61]. These protocols enable seamless communication and interoperability with SDIs, allowing GeoNLU to retrieve geospatial data from different sources.

4.3.2. Data source connectivity

GeoNLU provides built-in connectors and adapters for various data sources, including geographic information systems (GIS) [62], geospatial databases [63], and web services [64]. These connectors establish connections with the respective data sources, allowing GeoNLU to retrieve data efficiently. The connectors are designed to handle different data formats, such as shapefiles, GeoJSON, raster datasets, and relational databases, ensuring compatibility with diverse data sources.

4.3.3. Semantic annotation and ontologies

GeoNLU incorporates semantic web technologies, such as RDF (Resource Description Framework) [65] and OWL (Web Ontology Language) [66], for semantic annotation and ontology-based integration. By annotating geospatial data with standardized vocabularies and ontologies, GeoNLU enhances the interoperability and understanding of the data. This semantic annotation enables seamless integration and retrieval of relevant geospatial information based on the semantic

relationships captured in the data.

4.3.4. Spatial indexing and query optimization

To ensure efficient data retrieval, GeoNLU employs spatial indexing techniques. Spatial indexes, such as R-trees or Quad-trees, are utilized to organize geospatial data in a hierarchical structure, enabling faster spatial queries and reducing the search space. Additionally, GeoNLU incorporates query optimization strategies to minimize the response time and improve the performance of spatial queries. Techniques like cost-based optimization and query rewriting are employed to optimize query execution plans and enhance data retrieval efficiency [67,68].

4.3.5. Caching and data preprocessing

GeoNLU incorporates caching mechanisms to store frequently accessed geospatial data and query results. Caching reduces the response time for subsequent queries by retrieving data from the cache instead of accessing the data source. Furthermore, GeoNLU supports data preprocessing techniques, such as data aggregation, filtering, and transformation, to optimize data retrieval. Preprocessing operations help reduce the data volume and eliminate redundant or irrelevant information, improving the efficiency of data retrieval processes [69,70].

4.3.6. Extensibility and customization

GeoNLU is designed to be extensible and customizable to accommodate different SDIs and data sources. It provides APIs and interfaces that allow developers to integrate new connectors, protocols, or data formats based on specific requirements. This extensibility ensures that GeoNLU seamlessly integrates with a wide range of SDIs and data sources, accommodating different data models and standards.

By incorporating these integration strategies, GeoNLU ensures compatibility and efficient data retrieval from different SDIs and data sources. The standardized protocols, connectors, semantic annotation, spatial indexing, caching, and extensibility features enable seamless integration, enabling users to interact with geospatial data from diverse sources within a unified framework.

5. Query processing and analysis

The representation of the query processing and analysis is represented in Fig. 8.

5.1. Natural language query interpretation and understanding

Interpreting and understanding natural language queries in the geospatial context involves employing various techniques to extract relevant information and understand the spatial relationships and entities mentioned in the query. Here are some key techniques used in GeoNLU.

5.1.1. Parsing

Parsing is the process of analyzing the grammatical structure of a sentence to understand its syntactic components and relationships. In the context of natural language understanding, parsing is used to identify the syntactic structure of the query and extract essential elements such as nouns, verbs, adjectives, and prepositional phrases. This helps in identifying the key entities and actions mentioned in the query [71].

5.1.2. Semantic Role Labeling (SRL)

Semantic Role Labeling is a technique used to identify the semantic roles of words or phrases in a sentence. It aims to understand the underlying meaning of a sentence by labeling each word or phrase with its corresponding semantic role, such as agent, patient, location, time, or manner. In the geospatial context, SRL helps in identifying the spatial entities, spatial relations, and other relevant attributes mentioned in the query [72].

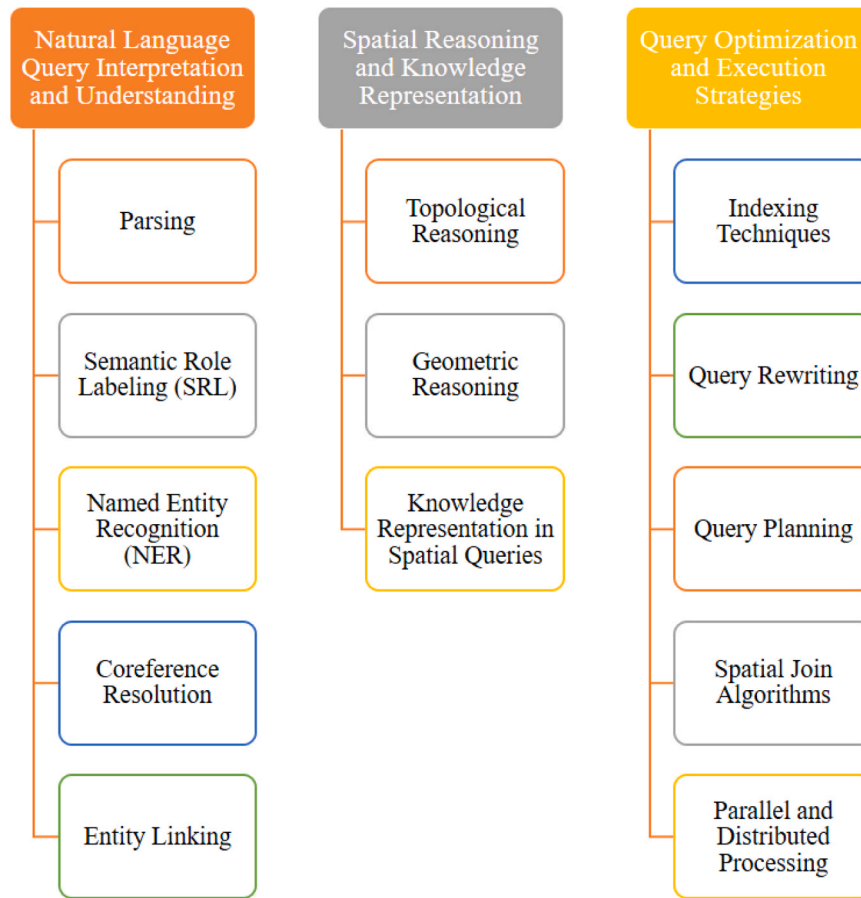


Fig. 8. Query Processing and Analysis.

5.1.3. Named Entity Recognition (NER)

Named Entity Recognition is a technique used to identify and classify named entities in text. In the geospatial domain, NER is employed to identify and extract specific geospatial entities from natural language queries, such as locations, addresses, landmarks, and geographic features. It helps in identifying the spatial context and entities mentioned in the query, enabling better understanding and accurate interpretation [73].

5.1.4. Coreference resolution

Coreference resolution is the process of determining when two or more expressions in a text refer to the same entity. In the context of geospatial NLU, coreference resolution helps in resolving pronouns or other referring expressions to their corresponding geospatial entities. This is crucial for understanding the spatial relationships and correctly interpreting the user's intent in the query [74].

5.1.5. Entity linking

Entity linking is the process of connecting mentions of named entities in text to their corresponding entries in a knowledge base or database. In the geospatial context, entity linking helps in mapping geospatial entities mentioned in the query to specific geographic locations or features. It aids in disambiguating similar entity mentions and ensures an accurate understanding of the geospatial context [75].

By employing parsing, semantic role labeling, named entity recognition, coreference resolution, and entity linking techniques, GeoNLU interpret and understand natural language queries in the geospatial context. These techniques facilitate the extraction of relevant spatial entities, spatial relationships, and other attributes from the query, enabling effective interaction with geospatial data based on natural

language inputs.

5.2. Spatial reasoning and knowledge representation

Spatial reasoning algorithms play a crucial role in GeoNLU by enabling the system to reason about spatial relationships and perform spatial operations on geospatial data. Two key types of spatial reasoning algorithms employed in GeoNLU are topological reasoning and geometric reasoning. Additionally, the importance of knowledge representation in spatial queries is emphasized to enhance the understanding and accuracy of the queries.

5.2.1. Topological reasoning

Topological reasoning focuses on reasoning about the topological relationships between spatial objects. It involves determining relationships such as containment, adjacency, intersection, and connectivity. Topological reasoning algorithms help in answering queries like "Which areas are adjacent to a given location?" or "Which features intersect a given area?" These algorithms utilize spatial indexes and data structures to determine topological relationships between spatial objects efficiently. By incorporating topological reasoning, GeoNLU accurately interpret and process queries that involve spatial relationships [76].

5.2.2. Geometric reasoning

Geometric reasoning involves reasoning about the geometric properties and measurements of spatial objects. It enables computations related to distances, angles, areas, and volumes. Geometric reasoning algorithms are utilized in GeoNLU to answer queries that involve geometric measurements, such as "What is the area covered by a given polygon?" or "What is the distance between two locations?" These

algorithms utilize geometric formulas and algorithms to perform accurate geometric calculations on geospatial data [77].

5.2.3. Knowledge representation in spatial queries

Knowledge representation plays a critical role in spatial queries as it provides a structured and standardized representation of geospatial data and its attributes. In GeoNLU, knowledge representation techniques, such as semantic web technologies (RDF, OWL), spatial ontologies, and spatial knowledge graphs, are employed to capture the spatial context and relationships of geospatial data. By representing spatial knowledge explicitly, GeoNLU enhances the understanding and accuracy of spatial queries. Knowledge representation enables more nuanced spatial reasoning by incorporating domain-specific knowledge and context, leading to more meaningful and precise query results [78].

The importance of knowledge representation in spatial queries lies in its ability to formally represent spatial entities, attributes, and relationships. Knowledge representation enables GeoNLU to reason about the semantics of spatial data and perform advanced spatial operations. Knowledge representation also facilitates the integration of heterogeneous geospatial data sources by providing a common representation framework. By leveraging knowledge representation, GeoNLU enhance spatial reasoning algorithms' accuracy, flexibility, and efficiency, leading to improved query results and a better understanding of the geospatial context.

The combination of topological reasoning, geometric reasoning, and knowledge representation in GeoNLU enables accurate interpretation of spatial relationships, precise geometric computations, and a comprehensive understanding of the spatial context, thereby enhancing the capabilities and effectiveness of spatial queries.

5.3. Query optimization and execution strategies

To optimize and execute geospatial queries efficiently, GeoNLU employs various methods and techniques. These techniques improve query performance, minimize response time, and reduce computational complexity. Here are some key methods used in GeoNLU.

5.3.1. Indexing techniques

GeoNLU utilizes indexing techniques to organize geospatial data efficiently and facilitate faster query processing. Spatial indexing structures, such as R-trees, Quad-trees, or Grid indexes, are employed to partition and organize spatial data in a hierarchical structure. These indexes enable efficient spatial query processing by reducing the search space and enabling faster data retrieval based on spatial relationships. By leveraging indexing techniques, GeoNLU significantly improve query performance, especially for spatial queries that involve large datasets [79].

5.3.2. Query rewriting

Query rewriting is a technique used to transform and optimize queries by modifying their structure or form. GeoNLU applies query rewriting in the geospatial context to transform complex or ambiguous queries into more efficient and precise forms. For example, it rewrite a complex spatial query involving multiple spatial relationships into a series of simpler queries or utilize spatial predicates to filter the search space before executing the query. Query rewriting helps optimize query execution plans and improves the efficiency of geospatial query processing [80].

5.3.3. Query planning

Query planning involves the selection of an optimal query execution plan based on cost estimation and query optimization techniques. In GeoNLU, query planning is performed to determine the most efficient approach for executing a geospatial query. It considers factors such as data distribution, available indexes, and the cost of different query execution strategies. By analyzing these factors, GeoNLU generate an

optimized query execution plan that minimizes the computational cost and maximizes query performance [81].

5.3.4. Spatial join algorithms

Spatial join algorithms are specifically designed to process queries that involve spatial relationships between multiple datasets. GeoNLU incorporates efficient spatial join algorithms, such as nested loop join, index-based join (e.g., R-tree-based join), or spatial hash join, to perform spatial joins between geospatial datasets. These algorithms optimize the join operations by exploiting spatial indexing and reducing the number of comparisons required, leading to faster query processing [82].

5.3.5. Parallel and distributed processing

To handle large-scale geospatial datasets and improve query performance, GeoNLU supports parallel and distributed processing techniques. By utilizing multiple computational resources, such as multi-core processors or distributed computing frameworks, GeoNLU execute geospatial queries in parallel, thereby reducing the overall response time. Parallel and distributed processing techniques enable GeoNLU to handle complex and resource-intensive geospatial queries efficiently [83].

By leveraging indexing techniques, query rewriting, query planning, spatial join algorithms, and parallel/distributed processing, GeoNLU optimizes and executes geospatial queries efficiently. These methods help minimize response time, reduce computational complexity, and improve query performance, enabling users to interact with geospatial data seamlessly and obtain query results in a timely manner.

6. Spatial knowledge graph

6.1. Construction and population of the knowledge graph

The process of constructing and populating a spatial knowledge graph involves several steps to integrate geospatial data sources, ontologies, and semantic annotations. The overview of the spatial knowledge graph is shown in Fig. 9.

6.1.1. Data source integration

The first step is to identify and integrate relevant geospatial data sources. These sources include GIS datasets, remote sensing data, sensor data, public datasets, or any other geospatial information repositories. The data sources are in various formats, such as vector data (e.g., points, lines, polygons) or raster data (e.g., satellite imagery). Integration involves data collection, cleaning, and transformation to ensure the consistency and compatibility of the data [84].

6.1.2. Ontology development

Next, an ontology specific to the domain of interest is developed. An ontology is a formal representation of concepts, relationships, and properties in a specific knowledge domain. In the context of spatial knowledge graphs, an ontology defines spatial entities, spatial relationships, attributes, and other relevant concepts. The ontology provides a common vocabulary and structure for organizing and representing geospatial knowledge [85].

6.1.3. Semantic annotation

Next, an ontology specific to the domain of interest is developed. An ontology is a formal representation of concepts, relationships, and properties in a specific knowledge domain. In the context of spatial knowledge graphs, an ontology defines spatial entities, spatial relationships, attributes, and other relevant concepts. The ontology provides a common vocabulary and structure for organizing and representing geospatial knowledge [86].

6.1.4. Knowledge graph construction

The constructed knowledge graph is a graph-based representation

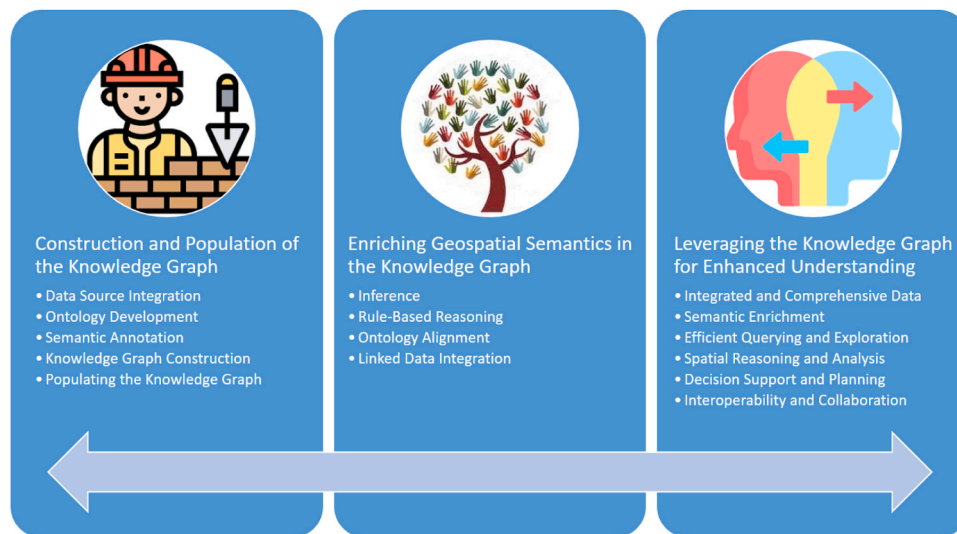


Fig. 9. Architecture of Spatial Knowledge Graph.

that captures the spatial entities, their attributes, and the relationships between them. The knowledge graph incorporates the geospatial data, ontological concepts, and semantic annotations. It represents the geospatial information in a structured and interconnected manner, enabling efficient navigation, querying, and inference. The knowledge graph is represented using semantic web technologies such as RDF (Resource Description Framework) and OWL (Web Ontology Language) [87].

6.1.5. Populating the knowledge graph

The populated knowledge graph is created by extracting and representing the geospatial data and their associated semantic annotations within the ontology-defined structure. This process involves mapping the data attributes to the corresponding ontology concepts, creating instances of the ontology classes, and establishing relationships between the entities based on the semantic annotations. The populated knowledge graph forms a rich and interconnected representation of geospatial information are queried, reasoned upon, and utilized for various applications [88].

By following these steps, the construction and population of a spatial knowledge graph enable the integration of geospatial data sources, ontologies, and semantic annotations. The resulting knowledge graph provides a unified and semantically-enriched representation of geospatial information, facilitating efficient querying, reasoning, and interoperability in the geospatial domain.

6.2. Enriching geospatial semantics in the knowledge graph

Enriching the spatial knowledge graph with additional semantic information involves techniques such as inference, rule-based reasoning, and ontology alignment. These techniques enhance the knowledge graph by deriving implicit knowledge, applying logical rules, and aligning with external ontologies.

6.2.1. Inference

Inference is the process of deriving implicit knowledge or new facts based on existing knowledge within the spatial knowledge graph. In the context of semantic information enrichment, inference is performed using semantic reasoning engines or logical frameworks such as Description Logic or Rule-Based Reasoning. Inference enables the discovery of hidden relationships, filling in missing information, and making logical deductions. For example, if the knowledge graph includes information about a river's source and mouth, inference is used to infer the length of the river based on the topological relationships [89].

6.2.2. Rule-based reasoning

Rule-based reasoning involves the application of logical rules or inference rules to make deductions or draw conclusions from the available information within the knowledge graph. These rules are defined using rule languages such as SWRL (Semantic Web Rule Language) or SPARQL Inferencing Notation (SPIN). Rule-based reasoning allows for implementing complex logical conditions and actions, enabling the discovery of new facts or the validation of existing facts. For example, a rule is defined to infer that if a region is designated as a protected area, then specific activities are prohibited within that region [90].

6.2.3. Ontology alignment

Ontology alignment is the process of establishing correspondences or mappings between concepts, relationships, or entities in different ontologies. By aligning the spatial knowledge graph with external ontologies or domain-specific ontologies, additional semantic information is integrated into the knowledge graph [91]. Ontology alignment techniques employ algorithms and matching methods to identify similar or equivalent concepts across ontologies and create mappings between them. This process helps bridge the gap between different ontologies and enriches the knowledge graph with additional semantic information from external sources [92].

6.2.4. Linked data integration

Linked Data integration involves linking the spatial knowledge graph with other datasets available on the web of data. By leveraging Linked Data principles and standards, such as RDF (Resource Description Framework), SPARQL (SPARQL Protocol and RDF Query Language), and URIs (Uniform Resource Identifiers), additional semantic information is integrated into the knowledge graph. Linked Data integration allows for the enrichment of the knowledge graph by connecting it with relevant and external datasets, expanding the scope and depth of the geospatial information [93].

By employing these techniques, the spatial knowledge graph are enriched with additional semantic information, expanding its knowledge representation capabilities and enhancing its utility. Inference, rule-based reasoning, ontology alignment, and Linked Data integration enable the discovery of implicit knowledge, logical deductions, and integration with external sources, making the knowledge graph more comprehensive and valuable for geospatial applications and analysis.

6.3. Leveraging the knowledge graph for enhanced understanding

Utilizing the spatial knowledge graph for geospatial analysis, exploration, and decision-making tasks offers several benefits.

6.3.1. Integrated and comprehensive data

The spatial knowledge graph integrates geospatial data from various sources, such as GIS datasets, sensor data, and public repositories. By bringing together diverse data sets into a unified framework, it provides a comprehensive view of geospatial information. This integrated data enables more holistic analysis and exploration of geospatial phenomena, allowing for a deeper understanding of spatial relationships and patterns [94].

6.3.2. Semantic enrichment

The semantic annotations and ontological representations within the spatial knowledge graph provide a rich and meaningful context to the geospatial data. This semantic enrichment enhances the understanding of the data, allowing for more accurate and nuanced analysis. The knowledge graph incorporating semantic information enables sophisticated queries, reasoning, and inference, leading to more insightful geospatial analysis [95].

6.3.3. Efficient querying and exploration

The spatial knowledge graph's structured and interconnected nature enables efficient querying and exploration of geospatial data. Users express complex queries using natural language or query languages like SPARQL, benefiting from the semantic understanding and spatial reasoning capabilities of the knowledge graph. This allows for faster retrieval of relevant information and supports interactive exploration of spatial relationships, facilitating decision-making processes [96].

6.3.4. Spatial reasoning and analysis

The spatial knowledge graph incorporates spatial reasoning techniques, such as topological and geometric reasoning, within its framework. This enables advanced spatial analysis tasks, including proximity analysis, spatial aggregation, spatial join operations, and spatial network analysis. By leveraging the spatial reasoning capabilities of the knowledge graph, users perform sophisticated spatial analysis tasks and derive meaningful insights from the data [97].

6.3.5. Decision support and planning

The spatial knowledge graph serves as a valuable resource for decision-making and planning activities. It provides a solid foundation for assessing the impacts of proposed changes, evaluating alternative scenarios, and analyzing spatial dependencies. With the knowledge graph, decision-makers access comprehensive and up-to-date geospatial information, enabling informed decisions and supporting effective planning strategies [98].

6.3.6. Interoperability and collaboration

The spatial knowledge graph promotes interoperability by providing a common framework for integrating and sharing geospatial data across different stakeholders and systems. It facilitates collaboration among researchers, policymakers, and domain experts by offering a unified representation of geospatial information. This interoperability encourages knowledge sharing, fosters interdisciplinary research, and enables collective decision-making processes [99].

By leveraging the benefits of the spatial knowledge graph, geospatial analysis, exploration, and decision-making tasks are enhanced. The integrated data, semantic enrichment, efficient querying, spatial reasoning, and decision support capabilities of the knowledge graph contribute to more insightful and effective geospatial analysis and decision-making processes.

7. User interface and interaction

7.1. Design principles for user-friendly geospatial NLU interfaces

Designing intuitive and user-friendly interfaces that facilitate natural language interaction with geospatial data requires careful consideration of user needs, context, and interaction patterns. Fig. 10 represents some best practices to follow.

7.1.1. Clear and concise language

The interface should use clear and concise language to ensure users understand the available functionalities and options. Avoid technical jargon and complex terms, opting for plain language that is easily understandable by users with varying levels of expertise [100].

7.1.2. Natural language input

Enable users to interact with the geospatial data using natural language input. Implement natural language processing (NLP) techniques to parse and understand user queries, allowing users to express their intent in a more conversational manner [101].

7.1.3. Contextual assistance

The interface should provide contextual assistance within the interface to guide users and offer suggestions. This includes auto-complete features, query suggestions, or inline help to assist users in formulating their queries and understanding the available options [102].

7.1.4. Feedback and validation

Provide immediate feedback to users to confirm the successful interpretation of their queries. If the query contains errors or ambiguity, provide suggestions or prompts to help users refine their queries. Clearly indicate any errors or invalid inputs and guide users towards valid alternatives [103].

7.1.5. Visualizations and maps

Use visualizations and interactive maps to present geospatial data in a visually appealing and intuitive manner. Visual representations help users understand spatial relationships, patterns, and trends more effectively. Enable users to interact with the visualizations, such as zooming, panning, and selecting specific areas of interest [104].

7.1.6. Progressive disclosure

Follow the principle of progressive disclosure, gradually presenting information and options as users interact. Start with a simple, focused interface and gradually reveal additional features and functionalities as users explore and express their needs [105].

7.1.7. Consistent and familiar design

Ensure consistency in the design and layout of the interface to create a familiar experience for users. Use standard design patterns and conventions, such as clear navigation menus, recognizable icons, and

User-Friendly Geospatial NLU Interfaces

- Clear and Concise Language
- Natural Language Input
- Contextual Assistance
- Feedback and Validation
- Visualizations and Maps
- Progressive Disclosure
- Consistent and Familiar Design
- Responsiveness and Performance
- User Testing and Iterative Design
- Accessibility Considerations

Fig. 10. Best practices for user-friendly Geospatial interfaces.

consistent terminology. This familiarity helps users navigate the interface more easily and reduces cognitive load [106].

7.1.8. Responsiveness and performance

Optimize the interface for responsiveness and performance, ensuring quick and smooth interactions. Slow response times or delays frustrate users and hinder their experience. Implement efficient algorithms, caching mechanisms, and proper data management techniques to deliver fast and seamless user interactions [107].

7.1.9. User testing and iterative design

Conduct user testing and gather feedback throughout the design and development process. Iterate the design based on user feedback, making improvements and refinements to enhance usability and user satisfaction. Regularly test the interface with representative users to identify pain points and address usability issues [108].

7.1.10. Accessibility considerations

Ensure the interface is accessible to users with diverse abilities. Follow accessibility guidelines and standards, including proper keyboard navigation, support for screen readers, and color contrast considerations. Design the interface to accommodate different devices and screen sizes [109].

By following these best practices, we create intuitive and user-friendly interfaces that enable natural language interaction with geospatial data. The goal is to provide a seamless and engaging experience for users, empowering them to explore and interact with geospatial information effortlessly.

7.2. Visualization techniques for presenting query results

When presenting geospatial query results, utilizing various visualization techniques effectively convey information and insights to users. The different visualization techniques commonly used in geospatial applications are represented in Fig. 11.

7.2.1. Maps

Maps are a fundamental visualization technique for geospatial data.



Fig. 11. Visualization Techniques for Presenting Query Results.

They provide a spatial context and allow users to visualize query results on a geographic representation. Maps include different types of information, such as points, lines, polygons, and choropleth maps. Interactive maps enable users to zoom in/out, pan, and click on features to explore details and better understand the spatial distribution of query results [110].

7.2.2. Charts and graphs

Charts and graphs are useful for visualizing geospatial query results that involve numerical or categorical data. Bar charts, pie charts, line charts, and scatter plots present statistical information associated with specific geographic regions or attributes. These visualizations help users identify patterns, trends, and relationships within the query results [111].

7.2.3. Heatmaps

Heatmaps represent the density or intensity of a particular attribute or event across a geographic area. They use color gradients to indicate the varying levels of intensity. Heatmaps are particularly effective for visualizing spatial patterns, such as population density, crime rates, or environmental factors, allowing users to quickly identify hotspots and variations across the map [112].

7.2.4. Proportional symbol maps

Proportional symbol maps use different-sized symbols to represent the magnitude or frequency of a specific attribute. For example, larger circles indicate higher values or frequencies, while smaller circles represent lower values or frequencies. This technique enables users to visually compare and analyze the distribution and concentration of query results across different geographic regions [113].

7.2.5. 3D visualization

In certain cases, 3D visualization techniques enhance the representation of geospatial query results, especially when dealing with elevation, terrain, or building models. By visualizing data in a three-dimensional space, users better understand the vertical dimension and perceive the relationships between objects or features [114].

7.2.6. Time animation

For geospatial query results that involve temporal aspects, time animation is employed. This technique shows the evolution of data over time by animating changes on the map. Visualizing data in a three-dimensional space allows users to better understand the vertical dimension and perceive the relationships between objects or features [115].

7.2.7. Interactive visualizations

Interactive visualizations engage users by providing interactive elements and controls. Users manipulate the visualization, filter data, change parameters, and explore different views. Interactive visualizations empower users to interact with the geospatial query results, enabling deeper exploration and analysis based on their specific interests and needs [116].

The choice of visualization technique depends on the nature of the geospatial query results, the type of data being presented, and the objectives of the analysis. Combining multiple visualization techniques provide a comprehensive understanding of the query results and facilitate effective communication of geospatial information to users.

7.3. Support for interactive exploration and refinement of queries

Interactive features play a crucial role in empowering users to refine and explore their queries in geospatial applications. Here are some interactive features commonly used to enhance query refinement and exploration.

7.3.1. Faceted search

Faceted search enables users to refine their queries by filtering results based on different facets or attributes. Facets represent different dimensions or characteristics of the data, such as location, time, category, or any other relevant attribute. Users select or deselect specific facets to narrow down the query results and focus on specific subsets of data. Faceted search provides an intuitive and flexible way for users to explore data from different perspectives and refine their queries iteratively [117].

7.3.2. Filtering

Filtering allows users to apply specific criteria or conditions to filter out unwanted data from query results. Users set filters based on attribute values, ranges, or other criteria to include or exclude specific data points. Filtering helps users focus on relevant information, refine their queries based on specific criteria, and investigate subsets of data that meet their specific requirements [118].

7.3.3. Dynamic query adjustments

Dynamic query adjustments allow users to modify their queries on-the-fly and observe the immediate impact on the query results. Users adjust parameters, such as spatial extent, time range, or attribute values, and instantly see the updated query results. This interactive feature enables users to iteratively explore different query options, fine-tune their queries, and gain a deeper understanding of the data [119].

7.3.4. Spatial interactions

Spatial interactions provide users with the ability to directly interact with the map or spatial representation of query results. Users select specific areas on the map, draw polygons or shapes to define spatial regions of interest or use spatial tools like buffers or proximity searches. Spatial interactions enable users to refine their queries based on spatial relationships, explore nearby features, or perform spatial analysis within the context of their queries [120].

7.3.5. Linked views

Linked views connect different visualizations or data representations to provide synchronized and coordinated exploration. Users interact with one visualization (e.g., a map) and observe the corresponding updates in other linked views (e.g., charts or tables). Linked views facilitate multidimensional exploration and enable users to simultaneously analyze query results from different angles, enhancing their understanding of the data [121].

7.3.6. Drill-down and drill-up

Drill-down and drill-up functionality allows users to navigate hierarchies or levels of detail in the data. Users drill down to explore more detailed information or drill up to see aggregated or summarized data. This interactive feature provides users with the flexibility to investigate different levels of granularity and gain insights at various levels of detail [122].

7.3.7. Visual query building

Visual query-building interfaces allow users to construct queries using visual elements and intuitive controls rather than writing complex queries manually. Users drag and drop elements, specify conditions, define spatial relationships, and set filters through a graphical interface. Visual query building simplifies the query creation process, making it accessible to users with varying levels of technical expertise [123].

By incorporating these interactive features, geospatial applications enable users to refine and explore their queries iteratively, providing a more engaging and efficient user experience. These features promote user-driven exploration, support data-driven decision-making, and facilitate the discovery of insights and patterns within geospatial data.

8. Applications and use cases

The integration of Geospatial Natural Language Processing (NLP) with Spatial Data Infrastructures (SDIs) has a wide range of applications and use cases across various domains. Here are some notable examples.

8.1. Urban planning and smart cities

GeoNLU are applied to urban planning and smart city initiatives. Users interact with geospatial data using natural language queries to analyze and understand urban environments. They explore factors like transportation infrastructure, land use patterns, population density, and environmental aspects to make informed decisions on urban development, resource allocation, and sustainable planning [124].

8.2. Environmental management

GeoNLU assist in environmental management by providing intuitive access to geospatial data related to environmental factors such as air quality, water resources, biodiversity, and land degradation. Users query and analyze this data to assess the impact of human activities on the environment, identify areas of concern, and support decision-making for conservation efforts and sustainable resource management [125].

8.2.1. Emergency management and disaster response

During emergencies or natural disasters, GeoNLU facilitate quick and efficient access to critical geospatial information. Users, including emergency responders and decision-makers, use natural language queries to gather real-time data on affected areas, infrastructure damage, evacuation routes, and resource allocation. This enables timely and informed decision-making for effective emergency response and disaster management [126].

8.2.2. Transportation and logistics

GeoNLU play a significant role in transportation and logistics planning. Users interact with geospatial data to analyze transportation networks, optimize routes, and identify potential congestion or bottlenecks. This information is used to improve transportation efficiency, enhance logistics operations, and support urban mobility planning.

8.2.3. Land management and real estate

In the context of land management and real estate, GeoNLU provide valuable insights for land administration, property valuation, and land use planning. Users query geospatial data to identify land ownership, zoning regulations, property boundaries, and other relevant information. This helps streamline land management processes, facilitate property transactions, and support urban development initiatives [127].

8.2.4. Natural resource exploration

GeoNLU aid in natural resource exploration and mining activities. Users query geospatial data to identify potential resource-rich areas, assess geological formations, and analyze environmental factors. This information enables more targeted exploration efforts, minimizes environmental impact, and supports sustainable resource extraction [128].

8.2.5. Tourism and cultural heritage

GeoNLU enhance tourism experiences by enabling users to explore and discover geospatial information about tourist attractions, landmarks, cultural heritage sites, and recreational areas. Users ask natural language queries to find nearby attractions, get information about historical sites, and plan their itineraries effectively [129].

These are just a few examples of how the integration of Geospatial NLP with SDIs are applied across different domains. The versatility of this technology allows for its adaptation to various use cases, promoting accessibility, democratizing geospatial information, and empowering

users to interact with geospatial data in a more intuitive and efficient manner.

8.3. Geospatial data exploration and discovery

GeoNLU enables users to explore and discover geospatial data through natural language queries, providing intuitive access to a wide range of geospatial information. Here's an illustration of how GeoNLU supports tasks such as data retrieval and filtering.

8.3.1. Data retrieval

Suppose a user would like to retrieve information about parks in a specific city using natural language. They input a query like, "Show me parks in New York City." GeoNLU processes the query and understands the user's intent to retrieve park information. It analyzes the query, identifies key entities like "parks" and "New York City," and translates the natural language query into a geospatial query. GeoNLU interacts with the underlying spatial data infrastructure and geospatial databases, using techniques like semantic annotation and ontology-based approaches to map the natural language query to relevant geospatial data sources. It retrieves the spatial data corresponding to parks in New York City and presents the results to the user meaningfully [130].

8.3.2. Filtering

GeoNLU also supports filtering geospatial data based on specific criteria provided in natural language queries. For example, a user might ask, "Show me parks in New York City with picnic areas." GeoNLU interprets the query, identifies the relevant entities and attributes, such as "parks," "New York City," and "picnic areas," and performs semantic annotation and entity recognition to understand the query's requirements. GeoNLU leverages the spatial knowledge graph and semantic understanding to filter the geospatial data accordingly. It applies filtering conditions based on the user's query, retrieving only the parks in New York City with picnic areas. The filtered results are presented to the user, providing them with targeted information that meets their specific criteria. By leveraging NLP techniques and the integration with SDIs, GeoNLU streamlines the process of exploring and discovering geospatial data. It eliminates users' need for specialized technical skills or knowledge of complex query languages. Users interact with geospatial data using natural language, making the process more accessible and intuitive [131].

GeoNLU not only retrieves relevant geospatial data based on natural language queries but also allows users to filter the data based on specific criteria. This capability empowers users to refine their search and explore geospatial information that aligns with their specific requirements, enhancing their ability to make informed decisions and gain valuable insights from the geospatial data available to them.

8.4. Decision support systems and spatial analytics

GeoNLU has significant potential in decision support systems and spatial analytics, offering powerful capabilities for tasks such as site selection, route optimization, and spatial trend analysis. Here are some examples showcasing the potential of GeoNLU in these domains: Site Selection: GeoNLU assist in site selection processes for various purposes, such as retail store locations, infrastructure development, or facility placement. Users interact with GeoNLU using natural language queries like "Find suitable locations for a new restaurant downtown" or "Identify potential sites for a solar power plant near residential areas." GeoNLU processes these queries, extracts relevant information, and combines it with geospatial data, including demographic data, land use patterns, transportation networks, and other relevant factors. By leveraging spatial reasoning algorithms, the system recommends potential sites that meet the specified criteria, enabling decision-makers to make informed choices in their site selection processes [132]. Route Optimization: GeoNLU optimize routes for various purposes, such as delivery logistics,

transportation planning, or emergency response. Users provide natural language queries like, "Find the shortest route for a truck to deliver goods to multiple locations" or "Identify the fastest route to reach a hospital from my current location." GeoNLU interprets these queries, understands the spatial context, and combines it with real-time data, including traffic conditions, road networks, and other relevant factors. The system employs spatial reasoning algorithms and optimization techniques to generate optimal routes that minimize travel time, distance, or other specified criteria. This enables efficient route planning and optimization in various applications [133].

Spatial Trend Analysis: GeoNLU support spatial trend analysis, allowing users to analyze patterns, correlations, and trends in geospatial data. For example, a user ask a query like, "Identify areas with increasing property prices in the last five years" or "Show me regions with high air pollution levels during winter months." GeoNLU processes these queries, identifies the relevant spatial and temporal components, and retrieves the corresponding geospatial data. It applies analytical techniques, such as clustering, regression analysis, or spatial interpolation, to detect data trends, patterns, and relationships. The system presents the results visually, enabling users to gain insights into spatial trends and make data-driven decisions [134]. These examples demonstrate the potential of GeoNLU in decision support systems and spatial analytics. By combining natural language interaction with geospatial data, GeoNLU enhances the accessibility and usability of spatial information, enabling users to perform complex spatial analyses and make informed decisions more efficiently.

8.5. Collaborative geospatial workflows and knowledge sharing

GeoNLU facilitates user collaboration and knowledge sharing by enabling easy sharing, annotation, and integration of geospatial information. Here are the key ways in which GeoNLU promotes collaboration and knowledge sharing.

8.5.1. Sharing geospatial information

GeoNLU allows users to share geospatial information with others seamlessly. Users not only retrieve and analyze geospatial data themselves but also share the results of their queries with colleagues or collaborators. This sharing is done through various channels, such as email, messaging platforms, or collaborative workspaces. By sharing geospatial information, users collaborate on common projects, exchange insights, and leverage each other's findings. This promotes collaboration and enables multiple stakeholders to access and benefit from the geospatial data and analyses performed using GeoNLU [135].

8.5.2. Annotation and contextual information

GeoNLU enables users to annotate geospatial data with contextual information and add relevant notes or comments. These annotations provide additional insights and interpretations that are shared with others. Users highlight specific features or areas of interest, add descriptions, or attach documents related to the geospatial data. Users provide valuable context and explanations by annotating geospatial information, facilitating better understanding and knowledge sharing among collaborators. Annotations serve as a means to communicate interpretations, findings, or recommendations, enhancing the collaborative nature of geospatial data exploration and analysis [136].

8.5.3. Integration of geospatial information

GeoNLU supports the integration of geospatial information from diverse sources and formats. Users integrate data from different datasets, spatial data infrastructures (SDIs), or external sources seamlessly. This integration enables users to combine and analyze data from multiple domains, enriching the overall knowledge and insights derived from geospatial information. By facilitating the integration of geospatial information, GeoNLU promotes knowledge sharing by providing a comprehensive view of spatial data. Users leverage the integrated data

to gain a holistic understanding, identify relationships, and discover new insights. This integration fosters collaboration by enabling users to work with a unified dataset and share their integrated findings with others [137].

8.5.4. Collaborative workspaces

GeoNLU provide collaborative workspaces where users share and collaborate on geospatial projects. These workspaces allow multiple users to access and interact with shared geospatial data, perform analyses, and contribute their insights. Users collaborate in real-time or asynchronously, depending on the nature of the project and the team's requirements. Collaborative workspaces facilitate knowledge sharing by creating a centralized platform for users to collaborate, discuss, and collectively work on geospatial tasks. They enable users to benefit from each other's expertise, contribute to ongoing projects, and build a collective knowledge base around geospatial information [138].

GeoNLU facilitates collaboration and knowledge sharing among users by enabling easy sharing of geospatial information, annotating data with contextual information, integrating diverse data sources, and providing collaborative workspaces. These features enhance the collaborative nature of geospatial data exploration, analysis, and decision-making, enabling users to leverage their peers' collective intelligence and expertise.

9. Challenges and future directions

9.1. Scalability and efficiency considerations

Scaling GeoNLU to handle large volumes of geospatial data pose several challenges. Here are some of the key challenges and potential solutions to address them.

9.1.1. Data storage and processing

Challenge. Large volumes of geospatial data require efficient storage and processing capabilities. Traditional storage and processing approaches cannot handle the scale and complexity of the data.

Solution. Employ scalable storage solutions, such as distributed file systems or cloud-based storage, that handle large geospatial datasets. Utilize distributed processing frameworks, such as Apache Spark or Hadoop, to parallelize and distribute geospatial data processing, allowing for faster and more efficient analysis.

9.1.2. Query performance

Challenge. Handling large volumes of geospatial data result in slow query performance and response times. As the dataset grows, the time taken to process queries and retrieve results become a bottleneck.

Solution. Implement indexing techniques, such as spatial indexing or quadtree-based indexing, to optimize query performance. These indexing methods enable faster spatial data retrieval and reduce the search space for queries. Additionally, consider using caching mechanisms to store frequently accessed data and pre-computed query results, reducing the need for repeated processing.

9.1.3. Computational resource management

Challenge. Scaling GeoNLU to handle large volumes of geospatial data requires efficient resource management. Allocating and managing computational resources, such as processing power and memory, becomes crucial to ensure optimal performance.

Solution. Utilize resource management frameworks, such as Apache Mesos or Kubernetes, to dynamically allocate and manage computational resources based on workload demands. These frameworks allow for efficient utilization of resources and scaling up or down based on the workload requirements.

9.1.4. Data integration and interoperability

Challenge. Integrating and harmonizing geospatial data from diverse

sources are challenging, especially when dealing with large volumes of data. Data formats, schema variations, and data quality issues hinder data integration and interoperability.

Solution. Develop data integration pipelines that handle diverse data formats and automate the data harmonization process. Implement data quality assessment and cleaning mechanisms to ensure the accuracy and consistency of the integrated geospatial data. To promote interoperability and seamless integration, utilize standardized data formats and schemas, such as GeoJSON or GeoSPARQL.

9.1.5. Scalable NLP techniques

Challenge. Scaling NLP techniques to handle large geospatial datasets requires efficient processing and understanding of natural language queries. Handling the linguistic complexities, disambiguation, and semantic understanding becomes more challenging with increasing data volumes.

Solution. Utilize distributed NLP frameworks, such as Apache OpenNLP or spaCy, that process natural language queries in parallel and handle large volumes of textual data efficiently. Employ machine learning-based approaches, such as deep learning models or pre-trained language models like BERT, to enhance the accuracy and scalability of NLP techniques.

9.1.6. System architecture and design

Challenge. Designing a scalable system architecture for GeoNLU requires careful consideration of distributed computing principles, fault tolerance, and load balancing to handle the increased demands of large geospatial datasets.

Solution. Adopt a microservices-based architecture that enables modular and scalable components. Utilize message queue systems, such as Apache Kafka, for asynchronous communication between system components. Employ load-balancing techniques to distribute the processing load across multiple instances or clusters.

Addressing these challenges requires a combination of technological advancements, efficient algorithms, and scalable infrastructure. By leveraging distributed computing, indexing techniques, resource management frameworks, and scalable NLP approaches, GeoNLU effectively handle the scaling requirements posed by large volumes of geospatial data.

9.2. Handling ambiguity and context in natural language queries

Ambiguity in natural language queries is a common challenge when integrating NLP techniques with geospatial data. Ambiguity arises due to the inherent flexibility and variability of human language. Disambiguating queries and achieving context-aware understanding are crucial for accurate interpretation and meaningful results. Some approaches to address the issue of ambiguity in natural language queries follow here.

9.2.1. Syntactic and semantic parsing

Syntactic and semantic parsing techniques aim to analyze the structure and meaning of natural language queries. Syntactic parsers analyze the grammatical structure of sentences, identifying the relationships between words and phrases. Semantic parsers go beyond syntax to extract the meaning of sentences, mapping them to formal representations such as semantic graphs or logical forms.

These parsing techniques help in disambiguating queries by identifying the syntactic and semantic roles of words and resolving ambiguities based on the context. Dependency parsing, constituency parsing, and semantic role labeling are some common approaches used for parsing natural language queries.

9.2.2. Named Entity Recognition (NER)

Named Entity Recognition is the task of identifying and classifying named entities, such as locations, organizations, or people, within a text.

NER plays a crucial role in disambiguating geospatial queries by recognizing and resolving references to specific locations or entities.

NER models trained on geospatial data accurately identify and disambiguate named entities in queries. The system better understand the context and provide relevant results by linking named entities to specific geospatial features or ontological entities.

9.2.3. Coreference resolution

Coreference resolution aims to identify and resolve references to entities that have been mentioned previously in the text. It helps disambiguate pronouns, definite and indefinite articles, and other expressions that refer to previously mentioned geospatial entities.

By resolving coreferences, the system link pronouns or other referring expressions to their corresponding entities, providing a more precise understanding of the query's context.

9.2.4. Contextual understanding and contextual word embeddings

Contextual understanding is crucial for disambiguating queries and capturing their intended meaning. Contextual word embeddings, such as ELMo, GPT, or BERT, capture the contextual information of words and phrases in a sentence.

These models generate word representations considering the surrounding context, enabling better disambiguation. By leveraging pre-trained contextual word embeddings, NLP systems better capture the nuances and meaning of geospatial queries.

9.2.5. Contextual knowledge and background knowledge

Leveraging contextual and background knowledge enhance the disambiguation process. Contextual knowledge includes information about the user's preferences, location, or historical context, which are used to disambiguate and personalize the query results.

Background knowledge, such as geospatial ontologies or knowledge graphs, provide additional contextual information to disambiguate queries. By incorporating domain-specific knowledge, NLP systems better understand the intended meaning of geospatial queries and provide more accurate results.

Combining these approaches, including syntactic and semantic parsing, named entity recognition, coreference resolution, contextual understanding, and contextual knowledge, help address the issue of ambiguity in natural language queries. By disambiguating queries and achieving context-aware understanding, NLP systems accurately interpret geospatial queries and provide relevant and meaningful results.

9.3. Privacy and security concerns in geospatial NLU

Geospatial NLU introduces several privacy and security challenges that need to be addressed to ensure the protection of sensitive geospatial data. Several key challenges associated with geospatial NLU in terms of privacy and security follow here.

9.3.1. Data anonymization

Geospatial data often contains personally identifiable information (PII), such as location traces or addresses. An important challenge is anonymizing the data to protect individuals' privacy while maintaining its usefulness for NLU tasks. Techniques like k-anonymity, differential privacy, or generalization are applied to de-identify geospatial data and prevent re-identification of individuals.

9.3.2. Access control

Controlling access to geospatial data is crucial to prevent unauthorized use or disclosure. Access control mechanisms need to be in place to ensure that only authorized individuals or systems access sensitive geospatial data. Role-based access control (RBAC), attribute-based access control (ABAC), or other access control models are employed to enforce security policies and restrict access based on user roles, privileges, or other attributes.

9.3.3. Secure query processing

Geospatial NLU involves processing queries that contain sensitive information. It is essential to ensure secure query processing to protect the confidentiality and integrity of the data. Encryption techniques, such as secure socket layer (SSL) or transport layer security (TLS), are employed to encrypt data transmission between the NLU system and the geospatial data sources, preventing unauthorized interception or tampering of queries and results.

9.3.4. Secure storage and data governance

Geospatial data used in NLU systems needs to be securely stored to prevent unauthorized access or data breaches. Robust security measures, including encryption at rest, access controls, and regular security audits, should be implemented to safeguard the stored data. Additionally, proper data governance practices, including data classification, data lifecycle management, and data retention policies, should be followed to ensure compliance with privacy regulations and minimize the risk of data breaches.

9.3.5. Privacy-preserving techniques

Privacy-preserving techniques are employed to enhance privacy while performing geospatial NLU tasks. Techniques such as secure multi-party computation (SMC), secure data aggregation, or homomorphic encryption enable computations to be performed on encrypted data, preserving privacy. These techniques allow for the analysis and processing of geospatial data while protecting sensitive information.

9.3.6. User consent and transparency

Transparency and user consent are important aspects of geospatial NLU systems. Users should be informed about the types of data being collected, how it will be used, and their rights regarding privacy. Providing clear privacy policies, obtaining explicit consent, and giving users control over their data help establish trust and ensure compliance with privacy regulations.

It is crucial for geospatial NLU systems to incorporate privacy and security considerations into their design and implementation. By employing techniques such as data anonymization, access control, secure query processing, privacy-preserving techniques, and ensuring user consent and transparency, the privacy and security challenges associated with geospatial NLU are effectively addressed, safeguarding sensitive geospatial data, and protecting individuals' privacy.

10. Conclusion

10.1. Recap of the GeoNLU framework

The GeoNLU framework is a cutting-edge solution that bridges the gap between natural language understanding (NLU) and spatial data infrastructures (SDIs). It enables seamless interaction with geospatial data using natural language queries, promoting accessibility and democratizing geospatial information. The framework incorporates state-of-the-art NLP techniques, such as semantic annotation, ontology-based approaches, and information extraction, to accurately interpret and understand geospatial queries. It leverages a spatial knowledge graph to enhance geospatial understanding and supports efficient query processing and visualization. With its modular design and integration with SDIs, GeoNLU revolutionizes geospatial data interaction, enabling users to effortlessly explore, analyze, and make informed decisions based on geospatial information.

10.2. Potential impact and benefits of GeoNLU

The potential impact of GeoNLU is profound. By bridging the gap between natural language understanding and spatial data infrastructures, it brings about transformative benefits. GeoNLU promotes accessibility and democratizes geospatial information, empowering a

broader range of users to interact with and understand complex geo-spatial data. It enhances user productivity by eliminating the need for specialized technical skills and complex query languages. GeoNLU enables faster and more intuitive data retrieval, exploration, and analysis, leading to improved decision-making in domains such as urban planning, environmental management, and transportation. Ultimately, GeoNLU accelerates geospatial insights, fosters collaboration, and unlocks the full potential of geospatial data in various fields, bringing about positive societal and economic impacts.

10.3. Future research directions and areas of improvement

Advancements in semantic parsing and knowledge representation improve the system's understanding of complex natural language queries and enable more accurate interpretation of spatial context. Extending GeoNLU to handle multiple languages will increase its applicability and usability worldwide. Incorporating contextual information, such as user preferences, historical data, and real-time events, enhance query interpretation and provide more personalized and relevant geospatial results. Further optimization techniques, such as distributed processing and parallel query execution, address the challenge of handling large volumes of geospatial data and improve system performance. Implementing mechanisms for real-time data integration and updates will ensure the freshness of geospatial information and support dynamic queries and analyses. Developing feedback mechanisms to learn from user interactions and adapt the system to user preferences and needs will enhance the user experience and improve the accuracy of query interpretation. Exploring integration with emerging technologies like machine learning, deep learning, and knowledge graph embeddings enhance the capabilities and accuracy of GeoNLU for geospatial understanding and analysis. By addressing these research directions and areas of improvement, GeoNLU continue to evolve, providing more sophisticated and user-friendly natural language interaction with geospatial data and unlocking new opportunities for geospatial analysis and decision-making.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors declare that they did not use AI-assisted technologies to create this article.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Pavel Trojovský reports financial support was provided by University of Hradec Králové, Czech Republic. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] S. Ghosh, A. Eldawy, AID*: a spatial index for visual exploration of geo-spatial data, *IEEE Trans. Knowl. Data Eng.* 34 (8) (2022) 3569–3582, <https://doi.org/10.1109/TKDE.2020.3026657>.
- [2] M. Tabatabaei, R. Kimiaefar, A. Hajian, et al., Robust outlier detection in geo-spatial data based on LOLIMOT and KNN search, *Earth Sci. Inf.* 14 (2021) 1065–1072, <https://doi.org/10.1007/s12145-021-00610-9>.
- [3] E. Luo, M. Kuffer, J. Wang, Urban poverty maps - From characterising deprivation using geo-spatial data to capturing deprivation from space, *Sustain. Cities Soc.* 84 (2022), 104033, <https://doi.org/10.1016/j.scs.2022.104033>.
- [4] A. Schwering, Approaches to semantic similarity measurement for geo-spatial data: a survey, *Trans. GIS* 12 (2008) 5–29, <https://doi.org/10.1111/j.1467-9671.2008.01084.x>.
- [5] U. Alganci, G.N. Kuru, I.Y. Algan, E. Sertel, Vineyard site suitability analysis by use of multicriteria approach applied on geo-spatial data, *Geocarto Int.* 34 (12) (2019) 1286–1299, <https://doi.org/10.1080/10106049.2018.1493156>.
- [6] Kim E.N., Tang I.T., Choi S.P., Yoon Y.H. An application of geo-spatial information system for optimum site selection in urban planning. In: *Proceedings of IGARSS '93 - IEEE International Geoscience and Remote Sensing Symposium*, Tokyo, Japan: IEEE; 1993, p. 2088–2090. doi: [10.1109/IGARSS.1993.322043](https://doi.org/10.1109/IGARSS.1993.322043).
- [7] V. Bandaru, R. Yaramasu, C. Jones, C.R. Izaurralde, A.R. Reddy, et al., Geo-CropSim: a geo-spatial crop simulation modeling framework for regional scale crop yield and water use assessment, *ISPRS J. Photogramm. Remote Sens.* 183 (2022) 34–53, <https://doi.org/10.1016/j.isprsjprs.2021.10.024>.
- [8] J. Huang, Y. Zhang, M. Deng, Z. He, Mining crowdsourced trajectory and geo-tagged data for spatial-semantic road map construction, *Trans. GIS* 26 (2022) 735–754, <https://doi.org/10.1111/tgis.12879>.
- [9] H.W. Britz, P. Webb, The effect of an intervention using GIS-generated geo-spatial data on the promotion of spatial cognition and spatial perspective taking in grade 11 learners, *South Afr. Geogr. J.* 98 (1) (2006) 182–193, <https://doi.org/10.1080/03736245.2014.977815>.
- [10] S. Laato, T. Tregel, Into the Unknown: Improving location-based gamified crowdsourcing solutions for geo data gathering, *Entertain. Comput.* 46 (2023), 100575, <https://doi.org/10.1016/j.entcom.2023.100575>.
- [11] X. Wu, Z. Cai, G. Wu, Q. Du, Integration and sharing of geo-spatial data based on data engine, *Geo-Spat. Inf. Sci.* 6 (4) (2003) 27–31, <https://doi.org/10.1007/BF02826946>.
- [12] Püerner H.A., Püerner B. Structured Query Language (SQL). In: *DB/2 kompakt. Vieweg+Teubner Verlag*, Wiesbaden, 1994. doi: [10.1007/978-3-322-83071-5_5](https://doi.org/10.1007/978-3-322-83071-5_5).
- [13] A. Borrmann, E. Rank, Specification and implementation of directional operators in a 3D spatial query language for building information models, *Adv. Eng. Inform.* 23 (1) (2009) 32–44, <https://doi.org/10.1016/j.aei.2008.06.005>.
- [14] U. Riebesell, A steep learning curve, *Nat. Geosci.* 6 (2013) 12–13, <https://doi.org/10.1038/ngeo1690>.
- [15] C. Wachter, Representing spatial concepts: modern east asian history in a digital publication format, *Hist. Theory* 61 (2022) 178–190, <https://doi.org/10.1111/hith.12285>.
- [16] M. Birjali, A. Beni-Hssane, M. Erritali, Evaluation of high-level query languages based on mapreduce in big data, *J. Big Data* 5 (2018) 36, <https://doi.org/10.1186/s40537-018-0146-3>.
- [17] A. Umair, E. Masciari, M.H. Ullah, Vaccine sentiment analysis using BERT + NBSVM and geo-spatial approaches, *J. Supercomput.* 79 (2023) 17355–17385, <https://doi.org/10.1007/s11227-023-05319-8>.
- [18] Yuan K., O'Neil P., Torrejon D. 8 - Landsat's past paves the way for data democratization in earth science. In: *Data Democracy*. Editor(s): Feras A. Batareseh, Ruixin Yang, Academic Press, 2020, 147–61. doi: [10.1016/B978-0-12-818366-3.00008-3](https://doi.org/10.1016/B978-0-12-818366-3.00008-3).
- [19] A. Järvelin, H. Keskustalo, E. Sormunen, M. Saastamoinen, K. Kettunen, Information retrieval from historical newspaper collections in highly inflectional languages, *J. Assn Inf. Sci. Tec.* 67 (2016) 2928–2946, <https://doi.org/10.1002/asi.23379>.
- [20] G. McDermott, C. Mulvihill, In-depth natural language understanding: a challenge for machine learning, *Ir. J. Psychol.* 10 (2) (1989) 291–303, <https://doi.org/10.1080/03033910.1989.10557748>.
- [21] Q. Lin, J. Zhao, B. Du, G. Fu, Z. Yuan, MEDNet: multiexpert detection network with unsupervised clustering of training samples, *IEEE Trans. Geosci. Remote Sens.* 60 (2022), 4703114, <https://doi.org/10.1109/TGRS.2021.3093556>.
- [22] L. Domegan, P. Garvey, P. McKeown, et al., Geocoding cryptosporidiosis cases in Ireland (2008–2017)—development of a reliable, reproducible, multiphase geocoding methodology, *Ir. J. Med. Sci.* 190 (2021) 1497–1507, <https://doi.org/10.1007/s11845-020-02468-0>.
- [23] Amar Ali-bey, Brahim Chaib-draa, Philippe Giguère, GSV-Cities: toward appropriate supervised visual place recognition, *ISSN 0925-2312, Neurocomputing Volume 513 (2022) 194–203*, <https://doi.org/10.1016/j.neucom.2022.09.127>.
- [24] G. Paltoglou, Sentiment-based event detection in Twitter, *J. Assn Inf. Sci. Tec.* 67 (2016) 1576–1587, <https://doi.org/10.1002/asi.23465>.
- [25] Veltmeijer E., Gerritsen E. SentiMap: Domain-Adaptive Geo-Spatial Sentiment Analysis. In: *2023 IEEE 17th International Conference on Semantic Computing (ICSC)*, Laguna Hills, CA, USA, 2023, 17–24. doi: [10.1109/ICSC56153.2023.00010](https://doi.org/10.1109/ICSC56153.2023.00010).
- [26] B.G. Banik, S.K. Bandyopadhyay, Novel text steganography using natural language processing and part-of-speech tagging, *IETE J. Res.* 66 (3) (2020) 384–395, <https://doi.org/10.1080/03772063.2018.1491807>.
- [27] H. Kang, J. Xiao, Y. Zhang, et al., A research toward chinese named entity recognition based on transfer learning, *Int J. Comput. Intell. Syst.* 16 (2023), 56, <https://doi.org/10.1007/s44196-023-00244-3>.
- [28] B. Wan, et al., A deep neural network model for coreference resolution in geological domain, *Inf. Process. Manag.* 60 (3) (2023), 103268, <https://doi.org/10.1016/j.ipm.2023.103268>.
- [29] M. Gattis, Mapping relational structure in spatial reasoning, *Cogn. Sci.* 28 (2004) 589–610, https://doi.org/10.1207/s15516709cog2804_4.

- [30] W. Qi, R. Procter, J. Zhang, W. Guo, Mapping consumer sentiment toward wireless services using geospatial twitter data, *IEEE Access* 7 (2019) 113726–113739, <https://doi.org/10.1109/ACCESS.2019.2935200>.
- [31] M. Yu, et al., Spatiotemporal event detection: a review, *Int. J. Digit. Earth* 2020 13 (12) (2020) 1339–1365, <https://doi.org/10.1080/17538947.2020.1738569>.
- [32] S.A. Crossley, K. Kyle, D.S. McNamara, Sentiment analysis and social cognition engine (SEANCE): an automatic tool for sentiment, social cognition, and social-order analysis, *Behav. Res* 49 (2017) 803–821, <https://doi.org/10.3758/s13428-016-0743-z>.
- [33] J. Cunha, R. Duarte, T. Guimarães, M.F. Santos, OpenEHR and Business Intelligence in healthcare: an overview, *Procedia Comput. Sci.* 220 (2023) 874–879, <https://doi.org/10.1016/j.procs.2023.03.118>.
- [34] K.R. Malik, M.A. Habib, S. Khalid, et al., A generic methodology for geo-related data semantic annotation, *Concurr. Comput. Pr. Exp.* 30 (2018), e4495, <https://doi.org/10.1002/cpe.4495>.
- [35] X. Wang, Q. Sun, J. Liang, JSON-LD based web API semantic annotation considering distributed knowledge, *IEEE Access* 8 (2020) 197203–197221, <https://doi.org/10.1109/ACCESS.2020.3034937>.
- [36] N. Chungoora, O. Canciglieri Jr., R.I.M. Young, Towards expressive ontology-based approaches to manufacturing knowledge representation and sharing, *Int. J. Comput. Integr. Manuf.* 23 (12) (2010) 1059–1070, <https://doi.org/10.1080/0951192X.2010.518976>.
- [37] P. Ayranci, P. Lai, N. Phan, et al., OnML: an ontology-based approach for interpretable machine learning, *J. Comb. Optim.* 44 (2022) 770–793, <https://doi.org/10.1007/s10878-022-00856-z>.
- [38] E. Zhu, et al., A unified framework of medical information annotation and extraction for Chinese clinical text, *Artif. Intell. Med.* 142 (2023), 102573, <https://doi.org/10.1016/j.artmed.2023.102573>.
- [39] N.W. Hickman, Implicit Knowledge, reasons, and semantic understanding, *Mind Lang.* 36 (2021) 707–728, <https://doi.org/10.1111/mila.12286>.
- [40] W.-K. Jia, R. R. Ying, X. Shi, Deploying a fast detection and eviction mechanism of invalid connection-oriented flow-entries in SDNs: a scalability approach, *IEEE Access* 8 (2020) 208669–208682, <https://doi.org/10.1109/ACCESS.2020.3036437>.
- [41] J. Lian, K. Naik, G.B. Agnew, L. Chen, M. Tamer Özsü, BBS: an energy efficient localized routing scheme for query processing in wireless sensor networks, *Int. J. Distrib. Sens. Netw.* 2 (1) (2006) 23–54, <https://doi.org/10.1080/15501320500330711>.
- [42] H. Le Sueur, I.N. Bruce, N. Geifman, et al., The challenges in data integration – heterogeneity and complexity in clinical trials and patient registries of Systemic Lupus Erythematosus, *BMC Med Res Method.* 20 (2020), 164, <https://doi.org/10.1186/s12874-020-01057-0>.
- [43] K. Han, R. Rezapour, K. Nakamura, D. Devkota, D.C. Miller, J. Diesner, An expert-in-the-loop method for domain-specific document categorization based on small training data, *J. Assoc. Inf. Sci. Technol.* 74 (6) (2023) 669–684, <https://doi.org/10.1002/asi.24714>.
- [44] M. Paul, L. Maglaras, M.A. Ferrag, I. Almomen, Digitization of healthcare sector: a study on privacy and security concerns, *ICT Express* 9 (4) (2023) 571–578, <https://doi.org/10.1016/j.icte.2023.02.007>.
- [45] W. Liu, W. Wu, Y. Wang, Y. Fu, Y. Li, Selective ensemble learning method for belief-rule-base classification system based on PAES, *Big Data Min. Anal.* 2 (4) (2019) 306–318, <https://doi.org/10.26599/BDMA.2019.9020008>.
- [46] M. Dvir, D. Ben-Zvi, Informal statistical models and modeling, *Math. Think. Learn.* 25 (1) (2023) 79–99, <https://doi.org/10.1080/10986065.2021.1925842>.
- [47] M. Tanveer, R. Rastogi, C.T. Lin, Advanced machine learning algorithms for biomedical data and imaging, *Multimed. Tools Appl.* 80 (2021) 30005, <https://doi.org/10.1007/s11042-021-11330-z>.
- [48] Buttenfield, B.P. (2002). Transmitting Vector Geospatial Data across the Internet. In: Egenhofer, M.J., Mark, D.M. (eds) *Geographic Information Science. GIScience 2002. Lecture Notes in Computer Science*, vol 2478. Springer, Berlin, Heidelberg. (https://doi.org/10.1007/3-540-45799-2_4).
- [49] I. Hong, A.T. Murray, Assessing raster GIS approximation for euclidean shortest path routing, *Trans. GIS* 20 (2016) 570–584, <https://doi.org/10.1111/tgis.12160>.
- [50] E. Roux, A.-P. Godillon-Maquinghen, P. Caulier, S. Bouilland, D. Bouttens, A support method for the contextual interpretation of biomechanical data, *IEEE Trans. Inf. Technol. Biomed.* 10 (1) (2006) 109–118, <https://doi.org/10.1109/TTB.2005.855566>.
- [51] G.U.O. Renzhong, Spatial objects and spatial relationships, *Geo-Spat. Inf. Sci.* 1998 1 (1) (1998) 38–42, <https://doi.org/10.1080/10095020.1998.10553282>.
- [52] X. Shuai, Z. Jing, H. Tuo, SAFuseNet: integration of fusion and detection for infrared and visible images, *AS* 5 (2022) 655–661, <https://doi.org/10.1007/s42401-022-00162-w>.
- [53] A. Al-Nazer, S. Albukhitan, T. Helmy, Cross-domain semantic web model for understanding multilingual natural language queries: English/Arabic health/food domain use case, 2016, *Procedia Comput. Sci.* 83 (2016) 607–614, <https://doi.org/10.1016/j.procs.2016.04.138>.
- [54] J. Laval, N. Amokrane, B. Thiam Niang, M. Derrass, N. Moalla, Data interoperability assessment, case of messaging-based data exchanges, *J. Softw. Evol. Proc.* (2023), e2538, <https://doi.org/10.1002/smr.2538>.
- [55] Amalia A., Afifa R.M., Herriyance H. Resource Description Framework Generation for Tropical Disease Using Web Scraping. In 2018 IEEE International Conference on Communication, Networks and Satellite (Comnetsat), Medan, Indonesia, 2018, 44–8, doi: [10.1109/COMNETSAT.2018.8684030](https://doi.org/10.1109/COMNETSAT.2018.8684030).
- [56] D. Paulraj, S. Swamynathan, M. Madhaiyan, Process model-based atomic service discovery and composition of composite semantic web services using web ontology language for services (OWL-S), *Enterp. Inf. Syst.* 6 (4) (2012) 445–471, <https://doi.org/10.1080/17517575.2011.654265>.
- [57] W. Ali, M. Saleem, B. Yao, et al., A survey of RDF stores & SPARQL engines for querying knowledge graphs, *Vldb J.* 31 (2022) 1–26, <https://doi.org/10.1007/s00778-021-00711-3>.
- [58] A.M. Castronova, J.L. Goodall, M.M. Elag, Models as web services using the open geospatial consortium (OGC) Web Processing Service (WPS) standard, *Environ. Model. Softw.* 41 (2013) 72–83, <https://doi.org/10.1016/j.envsoft.2012.11.010>.
- [59] G. Dong, R. Li, J. Jiang, H. Wu, S.C. McClure, Multigranular wavelet decomposition-based support vector regression and moving average method for service-time prediction on web map service platforms, *IEEE Syst. J.* 14 (3) (2020) 3653–3664, <https://doi.org/10.1109/JSYST.2019.2944527>.
- [60] C. Zhang, T. Zhao, W. Li, J.P. Osleeb, Towards logic-based geospatial feature discovery and integration using web feature service and geospatial semantic web, *Int. J. Geogr. Inf. Sci.* 24 (6) (2010) 903–923, <https://doi.org/10.1080/13658810903240687>.
- [61] He J., Sun W., Mi W., Jia K. A uniform approach for multi-version web coverage service retrieve based on dynamic schema matching. In 2011 International Conference on Remote Sensing, Environment and Transportation Engineering, Nanjing, 2011, p. 8032–36, doi: [10.1109/RSETE.2011.5964017](https://doi.org/10.1109/RSETE.2011.5964017).
- [62] K. Tzavella, A. Fekete, F. Friedrich, Opportunities provided by geographic information systems and volunteered geographic information for a timely emergency response during flood events in Cologne, Germany, *Nat. Hazards* 91 (1) (2018) 29–57, <https://doi.org/10.1007/s11069-017-3102-1>.
- [63] S. Kim, Y. Hoang, T.T. Yu, Y.S. Kanwar, GeoYCSB: a benchmark framework for the performance and scalability evaluation of geospatial NoSQL databases, *Big Data Res.* 31 (2023), 100368, <https://doi.org/10.1016/j.bdr.2023.100368>.
- [64] Yang C., Shao Y., N. Chen N., Di L. Aggregating distributed geo-processing workflows and web services as processing model web. In 2012 First International Conference on Agro-Geoinformatics (Agro-Geoinformatics), Shanghai, China, 2012, 1–4. doi: [10.1109/Agro-Geoinformatics.2012.6311638](https://doi.org/10.1109/Agro-Geoinformatics.2012.6311638).
- [65] W. Jiang, L. Yan, Y. Tu, X. Zhou, Z. Ma, PG-explorer: resource description framework data exploration with property graphs, *Expert Syst. Appl.* 198 (2022), 116789, <https://doi.org/10.1016/j.eswa.2022.116789>.
- [66] I. Horrocks, P.F. Patel-Schneider, F. van Harmelen, From SHIQ and RDF to OWL: the making of a web ontology language, *J. Web Semant.* 1 (1) (2003) 7–26, <https://doi.org/10.1016/j.websem.2003.07.001>.
- [67] C. Zhou, M. Li, Performance evaluation of spatial indexing to identify polygon intersection, *Geocarto Int.* 35 (16) (2020) 1850–1872, <https://doi.org/10.1080/10106049.2019.1624987>.
- [68] E. Azhir, N. Jafari Navimipour, M. Hosseinzadeh, A. Sharifi, A. Darwesh, Deterministic and non-deterministic query optimization techniques in the cloud computing, *Concurr. Comput. Pr. Exp.* 31 (2019), e5240, <https://doi.org/10.1002/cpe.5240>.
- [69] T.-V. Nguyen, N.-N. Dao, V. Dat Tuong, W. Noh, S. Cho, User-aware and flexible proactive caching using LSTM and ensemble learning in IoT-MEC networks, *IEEE Internet Things J.* 9 (5) (2022) 3251–3269, <https://doi.org/10.1109/JIOT.2021.3097768>.
- [70] A. Wurz, R. Manner, Flexible high-speed Fastbus master for data read-out and preprocessing, *IEEE Trans. Nucl. Sci.* 37 (2) (1990) 256–261, <https://doi.org/10.1109/23.106628>.
- [71] G.L. Brase, Using statistical reasoning performance to reveal information parsing preferences in the mind, *Q. J. Exp. Psychol.* 2015 68 (3) (2015) 459–472, <https://doi.org/10.1080/17470218.2014.944920>.
- [72] J.I. Butenko, M.L. Galetka, E.E. Sineva, Developing a Semantic Role Labeling System in Scientific and Technical Texts on Aviation and Astronautics, *Autom. Doc. Math. Linguist.* 56 (2022) 251–260, <https://doi.org/10.3103/S0005105522050077>.
- [73] R. Geng, Y. Chen, R. Huang, Y. Qin, Q. Zheng, Planarized sentence representation for nested named entity recognition, *Inf. Process. Manag.* 60 (4) (2023), 103352, <https://doi.org/10.1016/j.ipm.2023.103352>.
- [74] L. Márquez, M. Recasens, E. Sapena, Coreference resolution: an empirical study based on SemEval-2010 shared Task 1, *Lang. Resour. Eval.* 47 (2013) 661–694, <https://doi.org/10.1007/s10579-012-9194-z>.
- [75] W. Shen, J. Han, J. Wang, X. Yuan, Z. Yang, SHINE+: a general framework for domain-specific entity linking with heterogeneous information networks, *IEEE Trans. Knowl. Data Eng.* 30 (2) (2018) 353–366, <https://doi.org/10.1109/TKDE.2017.2730862>.
- [76] C. Bonura, Theorizing elsewhere: comparison and topological reasoning in political theory, *Polity* 45 (2013) 34–55, <https://doi.org/10.1057/pol.2012.31>.
- [77] J. Troup, H. Soto-Johnson, G. Karakok, et al., Developing students' geometric reasoning about the derivative of complex valued functions, *Digit. Exp. Math. Educ.* 3 (2017) 173–205, <https://doi.org/10.1007/s40751-017-0032-1>.
- [78] Y. Wang, L. Zhu, J. Ma, G. Wu, Hu, Jiangchuan Liu, Yaqiong Qiao, Knowledge graph-based spatial-aware user community preference query algorithm for LBSNs, *Big Data Res.* Volume 23 (2021), 100169, <https://doi.org/10.1016/j.bdr.2020.100169>.
- [79] S. Sharma, V. Gupta, M. Juneja, A survey of image data indexing techniques, *Artif. Intell. Rev.* 52 (2019) 1189–1266, <https://doi.org/10.1007/s10462-018-9673-8>.
- [80] M. Arenas, P. Barceló, R. Fagin, L. Libkin, Solutions and query rewriting in data exchange, *Inf. Comput.* 228–229 (2013) 28–61, <https://doi.org/10.1016/j.ic.2013.06.002>.
- [81] J.K. Levandoski, M.E. Khalefa, M.F. Mokbel, On producing high and early result throughput in multijoin query plans, *IEEE Trans. Knowl. Data Eng.* 23 (12) (2011) 1888–1902, <https://doi.org/10.1109/TKDE.2010.182>.

- [82] B. Qiao, B. Hu, J. Zhu, G. Wu, C. Giraud-Carrier, G. Wang, A top-k spatial join querying processing algorithm based on spark, *Inf. Syst.* 87 (2020), 101419, <https://doi.org/10.1016/j.is.2019.101419>.
- [83] C.R. Cox, M.S. Seidenberg, T.T. Rogers, Connecting functional brain imaging and parallel distributed processing, *Lang., Cogn. Neurosci.* 30 (4) (2015) 380–394, <https://doi.org/10.1080/23273798.2014.994010>.
- [84] X. Jia, Z. Cheng, B. Guo, Reliability analysis for system by transmitting, pooling and integrating multi-source data, *Reliab. Eng. Syst. Saf.* 224 (2022), 108471, <https://doi.org/10.1016/j.res.2022.108471>.
- [85] A. Smirnov, T. Levashova, A. Ponomarev, N. Shilov, Methodology for multi-aspect ontology development: ontology for decision support based on human-machine collective intelligence, *IEEE Access* 9 (2021) 135167–135185, <https://doi.org/10.1109/ACCESS.2021.3116870>.
- [86] P.S. Rani, R.M. Suresh, R. Sethukarasi, Multi-level semantic annotation and unified data integration using semantic web ontology in big data processing, *Clust. Comput.* 22 (5) (2019) 10401–10413, <https://doi.org/10.1007/s10586-017-1029-7>.
- [87] L. Ren, Y. Li, X. Wang, J. Cui, L. Zhang, An ABGE-aided manufacturing knowledge graph construction approach for heterogeneous IIoT data integration, *Int. J. Prod. Res.* 61 (12) (2023) 4102–4116, <https://doi.org/10.1080/00207543.2022.2042416>.
- [88] F.-L. Peng, Y.-K. Qiao, C. Yang, Building a knowledge graph for operational hazard management of utility tunnels, *Expert Syst. Appl.* 223 (2023), 119901, <https://doi.org/10.1016/j.eswa.2023.119901>.
- [89] Q. Guo, X. Wang, Z. Zhu, et al., A knowledge inference model for question answering on an incomplete knowledge graph, *Appl. Intell.* 53 (2023) 7634–7646, <https://doi.org/10.1007/s10489-022-03927-0>.
- [90] Li J., Sato A., Huang R., Chen D. A Rule-Based Knowledge Discovery Engine Embedded Semantic Graph Knowledge Repository for Retail Business. In 2016 International Conference on Advanced Cloud and Big Data (CBD), Chengdu, China, 2016, 81–86, doi: (0.1109/CBD.2016.024).
- [91] O. Sakka, P.-A. Millet, V. Botta-Genoulaz, An ontological approach for strategic alignment: a supply chain operations reference case study, *Int. J. Comput. Integr. Manuf.* 24 (11) (2011) 1022–1037, <https://doi.org/10.1080/0951192X.2011.575798>.
- [92] F. Yang, Variable weight semantic graph-based ontology mapping method, *Expert Syst.* 36 (2019), e12337, <https://doi.org/10.1111/exsy.12337>.
- [93] L. Zhao, R. Ichise, Ontology Integration for Linked Data, *J. Data Semant.* 3 (2014) 237–254, <https://doi.org/10.1007/s13740-014-0041-9>.
- [94] Wang G., Li T., Yue H., Yang Z., Zhang R. Integrating Heterogeneous Security Knowledge Sources for Comprehensive Security Analysis. In 2021 IEEE 45th Annual Computers, Software, and Applications Conference (COMPSAC), Madrid, Spain, 2021, 714–724. doi: (0.1109/COMPSAC51774.2021.00103).
- [95] H. Yin, J. Zhong, C. Wang, R. Li, X. Li, GS-InGAT: An interaction graph attention network with global semantic for knowledge graph completion, *Expert Syst. Appl.* 228 (2023), 120380, <https://doi.org/10.1016/j.eswa.2023.120380>.
- [96] C. Liu, X. Zhang, W. Huang, The exploration of objective task difficulty and domain knowledge effects on users' query formulation, *Proc. Assoc. Info Sci. Tech.* 53 (2016) 1–9, <https://doi.org/10.1002/pra2.2016.14505301063>.
- [97] B. Zhou, X. Shen, Y. Lu, X. Li, B. Hua, T. Liu, J. Bao, Semantic-aware event link reasoning over industrial knowledge graph embedding time series data, *Int. J. Prod. Res.* 61 (12) (2023) 4117–4134, <https://doi.org/10.1080/00207543.2021.2022803>.
- [98] Okamoto K., Tanaka H., Takemura T. et al. A hypothesis-generating support system using medical records for clinical knowledge acquisition. In The 6th International Conference on Soft Computing and Intelligent Systems, and The 13th International Symposium on Advanced Intelligence Systems, Kobe, Japan, 2012, 1130–1133. doi: (0.1109/SCIS-ISIS.2012.6505123).
- [99] F. Belkadi, N. Troussier, B. Eynard, et al., Collaboration based on product lifecycles interoperability for extended enterprise, *Int. J. Inter. Des. Manuf.* 4 (2010) 169–179, <https://doi.org/10.1007/s12008-010-0099-z>.
- [100] L.K. Fritz-Laylin, S.J. Lord, M. Kakley, R.D. Mullins, Concise language promotes clear thinking about cell shape and locomotion, *BioEssays* 40 (2018), 1700225, <https://doi.org/10.1002/bies.201700225>.
- [101] S. Saini, V. Sahula, Cognitive architecture for natural language comprehension, *Cogn. Comput. Syst.* 2 (2020) 23–31, <https://doi.org/10.1049/ccs.2019.0017>.
- [102] V.C. Hung, A.J. Gonzalez, Context-centric speech-based human-computer interaction, *Int. J. Intell. Syst.* 28 (2013) 1010–1037, <https://doi.org/10.1002/int.21614>.
- [103] A. Tarko, N.-D. Tsendbazar, S. de Bruin, A.K. Bregt, Producing consistent visually interpreted land cover reference data: learning from feedback, *Int. J. Digit. Earth* 2021 14 (1) (2021) 52–70, <https://doi.org/10.1080/17538947.2020.1729878>.
- [104] T. Vaittinen, D. McGooin, Uncover: supporting city exploration with egocentric visualizations of location-based content, *Pers. Ubiquit. Comput.* 22 (2018) 807–824, <https://doi.org/10.1007/s00779-018-1167-9>.
- [105] M.L. Howard, M.L. Gaviola, Progressive disclosure cases: the design and evaluation of use in multiple therapeutics courses, *Curr. Pharm. Teach. Learn.* 10 (6) (2018) 723–729, <https://doi.org/10.1016/j.cptl.2018.03.001>.
- [106] Z. Kakavand, A.A.N. Shirehjini, M.G. Moghaddam, S. Shirmohammadi, Child-home interaction: design and usability evaluation of a game-based end-user development for children, *Int. J. Child-Comput. Interact.* 37 (2023), 100594, <https://doi.org/10.1016/j.ijcci.2023.100594>.
- [107] I.B. Lima, W. Hwang, Effects of heuristic type, user interaction level, and evaluator's characteristics on usability metrics of augmented reality (AR) user interfaces, *Int. J. Hum. Interact.* (2023), <https://doi.org/10.1080/10447318.2022.2163769>.
- [108] Reckter H., Geiger C., Singer J., Streuber S. Tech-note: Iterative design and test of a multimodal experience. In 2009 IEEE Symposium on 3D User Interfaces, Lafayette, LA, USA, 2009, 99–102. doi: (0.1109/3DUI.2009.4811213).
- [109] Priegnitz D. et al. Human factors methods in the design of the graphical user interface for the open systems radar product generation (ORPG) component of the WSR-88D. In Proceedings of the IEEE 1997 National Aerospace and Electronics Conference. NAECON 1997, Dayton, OH, USA, 1997, 328–331. doi: (0.1109/NAECON.1997.618099).
- [110] Crowder T.J. Maps and Sets. In JavaScript®, T.J. Crowder (Ed.), 2020. doi: (0.1002/9781119367901.ch12).
- [111] J.D. Camm, G.E. McCray, M.L. Roehm, More than just charts and graphs: What to teach in a data visualization course, *Decis. Sci. J. Innov. Educ.* 21 (3) (2022) 112–122, <https://doi.org/10.1111/dsji.12282>.
- [112] I. Fekete, G. Hentschel, Interactive heatmaps as an improved means of analysing complex socio-dialectal patterns: german loans in silesian, *J. Quant. Linguist.* 2022 29 (4) (2022) 425–449, <https://doi.org/10.1080/09296174.2021.1898089>.
- [113] R.G. Cano, G. Kunigami, C.C. de Souza, P.J. de Rezende, A hybrid GRASP heuristic to construct effective drawings of proportional symbol maps, *Comput. Oper. Res.* 40 (5) (2013) 1435–1447, <https://doi.org/10.1016/j.cor.2012.09.007>.
- [114] L. Guo, et al., Fan, VT3D: a visualization toolbox for 3D transcriptomic data, *J. Genet. Genom.* 50 (9) (2023) 713–719, <https://doi.org/10.1016/j.jgg.2023.04.001>.
- [115] S.B. Steinman, M. Nawrot, Real-time color-frame animation for visual psychophysics on the Macintosh computer, *Behav. Res. Methods, Instrum., Comput.* 24 (1992) 439–452, <https://doi.org/10.3758/BF03203578>.
- [116] Schulz M., Reck F., Bertelheimer W., Ertl T. Interactive visualization of fluid dynamics simulations in locally refined cartesian grids. In Proceedings Visualization '99 (Cat. No.99CB37067), San Francisco, CA, USA, 1999, 413–553. doi: (0.1109/VISUAL.1999.809918).
- [117] K. McGrath, Musings on faceted search, metadata, and library discovery interfaces, *Cat. Classif. Q.* 2023 61 (5–6) (2023), <https://doi.org/10.1080/01639374.2023.2222120>.
- [118] S. Liu, C. Zhang, Personalized file data query matching method based on SOA, *Int. J. Comput. Appl.* 43 (2) (2021) 159–164, <https://doi.org/10.1080/1206212X.2018.1534374>.
- [119] A. Rahal, Q. Zhu, P.Å. Larson, Evolutionary techniques for updating query cost models in a dynamic multidatabase environment, *Vldb* 13 (2004) 162–176, <https://doi.org/10.1007/s00778-003-0110-4>.
- [120] N. Chen, X. Jin, Cross-industry asset allocation with the spatial interaction on multiple risk transmission channels, *North Am. J. Econ. Financ.* 67 (2023), 101935, <https://doi.org/10.1016/j.najef.2023.101935>.
- [121] J. Hwang, S. Park, LIDAB: a user-friendly display system for linked multimedia data and its application in education, *Multimed. Tools Appl.* 75 (2016) 13149–13162, <https://doi.org/10.1007/s11042-015-2994-6>.
- [122] V. da Silva, C. Esperança, R. Marroquim, OMICron – oblique multipass hierarchy creation while navigating, *Comput. Graph.* 84 (2019) 42–54, <https://doi.org/10.1016/j.cag.2019.08.016>.
- [123] T. Catarci, S.K. Chang, G. Santucci, Query representation and management in a multiparadigmatic visual query environment, *J. Intell. Inf. Syst.* 3 (1994) 299–330, <https://doi.org/10.1007/BF00962240>.
- [124] M. Thite, Smart cities: implications of urban planning for human resource development, *Hum. Resour. Dev. Int.* 145 (2011) 623–631, <https://doi.org/10.1080/13678868.2011.618349>.
- [125] M. Arashpour, AI explainability framework for environmental management research, *J. Environ. Manag.* 342 (2023), 118149, <https://doi.org/10.1016/j.jenvman.2023.118149>.
- [126] S. Andrews, H. Wang, D. Ni, S. Gao, J. Collura, Development and implementation of an adapted evacuation planning methodology in the framework of emergency management and disaster response: a case study using TransCAD, *J. Transp. Saf. Secur.* 2 (4) (2010) 352–368, <https://doi.org/10.1080/19439962.2010.517743>.
- [127] M. Hofmann, N.D. Müller, C.J. Stankiewicz, A. Pfnür, H.J. Linke, The effects of knowledge orders on climate change policy in urban land management and real estate management: a case study of three German cities, *Urban Res. Pract.* 2015 8 (3) (2015) 336–353, <https://doi.org/10.1080/17535069.2015.1051381>.
- [128] J. Allan, R. Ojeda-García, Natural resource exploitation in Western Sahara: new research directions, *J. North Afr. Stud.* 27 (6) (2022) 1107–1136, <https://doi.org/10.1080/13629387.2021.1917120>.
- [129] I. Bucurescu, Managing tourism and cultural heritage in historic towns: examples from Romania, *J. Herit. Tour.* 10 (3) (2015) 248–262, <https://doi.org/10.1080/1743873X.2014.968162>.
- [130] P. Meesad, Thai fake news detection based on information retrieval, natural language processing and machine learning, *SN Comput. Sci.* 2 (2021), 425, <https://doi.org/10.1007/s42979-021-00775-6>.
- [131] Y. Li, X. Deng, S. Ba, W.R. Myers, W.A. Brennenman, S.J. Lange, R. Zink, J. Ran, Cluster-based data filtering for manufacturing big data systems, *J. Qual. Technol.* 54 (3) (2022) 290–302, <https://doi.org/10.1080/00224065.2021.1889420>.
- [132] Y. Xue, C. Qu, M. Su, D. Qiu, Z. Li, X. Zhang, X. Ma, Subsea tunnel site selection evaluation: a new evaluation model, *Mar. Georesources Geotechnol.* 38 (10) (2020) 1187–1198, <https://doi.org/10.1080/1064119X.2019.1657210>.
- [133] C. Zhang, L. Cheng, Y. Zhang, M. Du, W. Zheng, Route optimization of vacant taxicab considering sequential dependence in abstract grid network based on quadtree, *Transp. Lett.* (2023), <https://doi.org/10.1080/19427867.2023.2193443>.
- [134] A. Malik, A. Kumar, P. Guhathakurta, et al., Spatial-temporal trend analysis of seasonal and annual rainfall (1966–2015) using innovative trend analysis method

- with significance test, Arab J. Geosci. 12 (2019), 328, <https://doi.org/10.1007/s12517-019-4454-5>.
- [135] J. Chen, L. Xiang, J. Gong, Virtual globe-based integration and sharing service method of GeoSpatial Information, Sci. China Earth Sci. 56 (2013) 1780–1790, <https://doi.org/10.1007/s11430-013-4627-0>.
- [136] T. Zhu, A.E.W. Johnson, J. Behar, et al., Crowd-sourced annotation of ECG signals using contextual information, Ann. Biomed. Eng. 42 (2014) 871–884, <https://doi.org/10.1007/s10439-013-0964-6>.
- [137] H. Meda, C. Vogiatzis, L.B. Davis, Integrating geospatial information in the analysis of network disruptions, Int. J. Disaster Risk Reduct. 87 (2023), 103569, <https://doi.org/10.1016/j.ijdr.2023.103569>.
- [138] C. Rodríguez-Domínguez, A.B. Márquez-Ruiz, J.L. Garrido, K. Benghazi, L. González-Herrera, A. Valenzuela, Technical note: a mobile collaborative workspace to assist forensic experts in disaster victim identification scenarios, Forensic Sci. Int 344 (2023), 111601, <https://doi.org/10.1016/j.forsciint.2023.111601>.